muchísimocks: An emulated cosmological simulation suite for simulation-based inference

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ABSTRACT

ksf: NOTE TO COAUTHORS (hi!): please define a macro with your initials and favorite color and comment in tex, do not use overleaf comment feature! thanks :)

Galaxy clustering contains rich cosmological information down to nonlinear scales; however, methods to extract this information are currently limited *dsj:* by the high cost of cosmological N-body simulations and by the fidelity of galaxy bias modeling. We develop a new approach to generating approximate yet robust galaxy mocks: we employ the `map2map` full-field emulator to generate particle displacement fields, and the hybrid Lagrangian bias expansion approach to flexibly model biased tracers. We create a suite of $(1\ h^{-1}\mathrm{Gpc})^3$ second-order bias fields at $z = 0$ spanning 10,000 different cosmologies, and weight these with multiple bias and noise parameter sets, resulting in 320,000 mock 3D *dsj:* galaxy density fields. We train a simulation-based inference (SBI) model on the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter power spectrum $P_{gm}(k)$, and the galaxy bispectrum $B_{ggg}(k_1, k_2, k_3)$. To test generalizability, we apply it to an independent N-body simulation populated with LRG-like galaxies using extended SubHalo Abundance Matching, and achieve unbiased cosmological constraints. Our results show that $P_{gm}(k)$ is critical for breaking parameter degeneracies, and that including $B_{ggg}(k_1, k_2, k_3)$ improves precision, on Ω_m by 12%, σ_8 by 11% and b_1 by 70%. In convergence tests, we find that 20,000 mocks are sufficient for $P_{gg}(k)$, beyond 320,000 are necessary for $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$, and that additional bias parameter sets provide similar gain as additional cosmologies *dsj:* gains from additional samples of bias parameters are comparable to gains from additional cosmologies? *ksf:* yes, beyond a few thousand cosmologies, is our finding somewhat surprisingly! hard to capture in a short way in the abstract, open to suggestions *dsj:* I agree, its tricky to phrase concisely and clearly, what about: and that additional bias parameter samples yield gains similar to those from additional cosmologies.. These findings are useful for guiding future SBI approaches for galaxy clustering, and demonstrate a promising path towards optimal inference and many other potential applications. The `muchísimocks` suite is available at `thisurl`. *ksf:* note: if we use only top model instead of ensemble, the gain from Bggg jumps to 39% for sigma8, which is significantly different. let's discuss using the top-3 ensemble vs just top model *ksf:* right around 250-word limit

Key words: methods: numerical – methods: statistical – galaxies: statistics – cosmology: large-scale structure of Universe – cosmological parameters

1 INTRODUCTION

The clustering of galaxies encodes information about the cosmological model describing the fundamental components of the universe. Surveys such as the Sloan Digital Sky Survey (York et al. 2000) and most recently the Dark Energy Spectroscopic Instrument (DESI, DESI Collaboration et al. 2023) have spectroscopically observed millions of galaxies, producing detailed maps of a large volume of the observable universe. Current and upcoming surveys such as Euclid (Euclid Collaboration et al. 2025) and the Roman Observatory (Spergel et al. 2015; Wang et al. 2022) will continue to deliver competitive and complementary galaxy clustering data sets.

Standard approaches to analyzing this spectroscopic survey data rely on analytic techniques to model summary statistics of the galaxy field, and Monte Carlo methods to sample the parameters of interest (e.g. Alam et al. 2017; Philcox & Ivanov 2022). Analyses are thus largely restricted to two-point *dsj:* and three-point (since you metion the bispectrum below) *ksf:* rearranged to avoid this issue, good point! statistics on quasi-linear scales, which contain robust cosmological information but do not fully capture the information content in the data. For instance, the DESI analyses use a perturbation theory approach, and their analysis is limited to $k_\mathrm{max} = 0.2$ for the power spectrum (Adame et al. 2025). **While higher order statistics such**

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as the bispectrum are now being incorporated, they require limiting to even larger scales given modeling difficulties; the DESI bispectrum analysis uses $k_{\rm max} = 0.12$ (Novell-Masot et al. 2025). Further, as surveys become more sensitive and statistical uncertainties improve, systematic errors will become increasingly relevant, and the effect of various systematics on clustering statistics—both two-point and higher-order statistics—is difficult to model.

A new approach known as simulation-based inference (SBI) has shown promise for cosmological inference from galaxy clustering, as it obviates these limitations *mpi:* which limitations? If you mean the modelling of observational systematics, you still need to model them in SBI as well, maybe if you specify which systematics you referr to in the previous sentence, this is ok. In the SBI approach, only a forward model of the data is required, and the posterior of the input parameters is approximated conditioned on the observed data. In the case of galaxy clustering, one can thus forward model the full galaxy field and directly measure any statistic, without the need for analytic modeling. Similarly, one can directly forward model the effects of systematics on the galaxy field, and the statistics can be simply measured without the need for additional modeling specific to each statistic. Finally, Monte Carlo methods typically assume a Gaussian form for the likelihood (and must estimate a nontrivial covariance matrix), even though higher-order statistics in particular are known to have non-Gaussian properties *mpi:* do you mean non-gaussian likelihood distributions or other properties as well? (Hahn et al. 2018); SBI makes no assumptions about the form of the likelihood. Further, in the case of galaxy clustering one can learn optimized statistics of the galaxy field for the parameters of interest, effectively using the full information in the galaxy field *mpi:* any citation here that you know of?.

While promising, SBI faces challenges for galaxy clustering analyses. One key issue is that SBI typically requires many simulations, but cosmological simulations are very computationally expensive. An N-body simulation with the volume and resolution to accurately describe galaxy clustering into the nonlinear regime costs at least hundreds (and up to millions) of CPU hours, and thousands to hundreds of thousands of simulations are generally needed for SBI. Another challenge is modeling how galaxies are connected to the underlying dark matter distribution, which involves complex and not yet well-understood galaxy formation physics; to be useful for SBI, then, models must be relatively fast, flexible enough to model various galaxy populations, and accurate to as small scales as possible.

There has been significant recent work that has demonstrated SBI as a promising path forward for the field. The `SimBIG` collaboration employs a forward model based on the 2,000 $(1h^{-1}\mathrm{Gpc})^3$ simulations of the Quijote Latin hypercube (Villaescusa-Navarro et al. 2020); they use the halo occupation distribution (HOD) model to populate the simulations with galaxies, choosing 10 HODs per cosmology for a total of 20,000 simulations (Hahn et al. 2023c). Using the power spectrum and applying to SDSS BOSS data (Dawson et al. 2016), they achieved constraints on σ_8 that are 27% more precise than the standard analysis thanks to the information in small scales (Hahn et al. 2022); subsequent analyses have applied `SimBIG` to the bispectrum (Hahn et al. 2023a), the marked power spectrum (Massara et al. 2024), and the full galaxy field (Lemos et al. 2023; Hahn et al. 2023b). Tucci & Schmidt (2024) take the effective field theory of large-scale structure (EFTofLSS) approach, generating 10^5 simulations and employing neural posterior estimation using the power spectrum and bispectrum out to $k_{\rm max} = 0.2$; as this model is fully perturbative, they are able to consider a Euclid-size volume of 63 $(h^{-1}\mathrm{Gpc})^3$. By performing a direct comparison to standard inference, they find that the off-diagonal elements of the covariance matrix are important for constraining σ_8 accurately, while allowing for a non-Gaussian likelihood is less important. Using the same forward model, Nguyen et al. (2024) show that full-field inference yields over a factor of five improvement on the constraint on σ_8 compared to the power spectrum and bispectrum at the same scale cut of $k_{\rm max} = 0.12$. Recent work has also explored forward models based on more physically motivated models of galaxy formation, albeit at the expense of much smaller volumes. de Santi et al. (2023) employ the 1,000-simulation Latin hypercubes of the CAMELS suite (Villaescusa-Navarro et al. 2021), which are hydrodynamical simulations with $(25\,h^{-1}\mathrm{Mpc})^3$ volumes, to perform SBI using graph neural networks; they find that their inference on $\Omega_{\rm m}$ is robust to galaxy formation physics.

Analyses that rely on N-body simulations, as some of these SBI approaches do, are limited by the high computational cost of these simulations. Significant effort has been spent developing *emulators* that learn the mapping from cosmological parameters to statistics of the density field, trained on a relatively small set of cosmologies and then able to effectively interpolate between them. Recently, full-field emulators have been developed that go beyond summary statistics, instead learning the full particle displacement field. The emulator `map2map` (Jamieson et al. 2023) uses a convolutional neural network-based architecture to map from Zel'dovich approximation displacements to N-body displacements, trained on the Quijote simulation suite (Villaescusa-Navarro et al. 2020). It has been used in a range of applications, such as producing large-volume emulated simulations (Scoggins et al. 2025); variants of `map2map` have been designed to emulate modified gravity (Saadeh et al. 2025) and to learn the mapping from nonlinear structures to the initial conditions of the universe (Doeser et al. 2024). There has been continued work on developing full-field emulators, with a range of training data and neural architectures (Kang et al. 2026; Ullmo et al. 2026; Jacobus et al. 2026).

Robustly modeling the galaxy–dark matter connection is a key challenge (Wechsler & Tinker 2018). Halo-based models such as the halo occupation distribution (e.g. Peacock & Smith 2000; Seljak 2000; Zheng et al. 2005) populate dark matter halos with galaxies based on halo mass and properties, but their flexibility degrades their robustness, particularly for higher-order statistics (e.g. Contreras et al. 2024; Wang et al. 2026). More physically motivated models like subhalo-abundance matching (e.g. Vale & Ostriker 2004; Shankar et al. 2006; Conroy et al. 2006) are more robust, but require merger trees which are more computationally expensive; even more complicated models like semi-analytic models (e.g. Somerville et al. 2008; Henriques et al. 2015; Stevens et al. 2016; Somerville & Davé 2015) and hydrodynamical simulations (e.g. Springel et al. 2005; Vogelsberger et al. 2014; Schaye et al. 2015) are both expensive and make specific choices regarding galaxy formation, which limits their generalizability. EFTofLSS offers a perturbative approach that is robust on large scales, but breaks down on small scales (Bernardeau et al. 2002; Desjacques et al. 2017).

The hybrid Lagrangian bias expansion (or hybrid effective field theory) offers an approach that is accurate into the small-scale regime while remaining computationally relatively low-cost (Modi et al. 2020; Kokron et al. 2021; Zennaro et al. 2023). It combines the Lagrangian bias expansion, which models the galaxy field as a summation of operators of the dark matter field, and for the displacement field uses uses particle displacements from N-body simulations rather than EFT's perturbative approach. This approach has been shown to be effective down to $k \sim 0.6$ *mpi:* units, and has been used to construct accurate emulators of the power spectrum (Pellejero-Ibanez et al. 2022, 2023b) *mpi:* The way I understand this is that the Pellejero-Ibanez 2022 should go to the place where you

have Modi, and Kokron, and then move Zennaro here. Pellejero-Ibanez et al 2022 is not the presentation of an emulator, it is the presentation of the HEFT with RSDs Also Pellejero-Ibanez 2023b should be 2024b. Pellejero Ibanez et al. (2023a) *mpi:* shouldn't this be Pellejero-Ibanez et al 2024a? showed that emulated N-body displacements with `map2map` can be used as inputs to the hybrid bias expansion to provide an accurate model of galaxy abundance. They found 1-2% precision on the power spectrum down to $k\sim0.6$ *dsj:* units; while there were some biases at intermediate scales, these were absorbed by nuisance parameters describing stochasticity. Notably, they found that the robustness held with variations in the cosmological parameters, demonstrating promise for using this approach for cosmological inference.

The combination of emulated N-body displacements and the hybrid bias expansion creates the opportunity for cosmological inference using the hybrid bias expansion for statistics beyond the power spectrum for the first time. The hybrid bias expansion has been thoroughly validated and applied for cosmological inference from the power spectrum, and its balance of robustness and speed makes it ideal for inference compared to other galaxy–halo connection approaches. With the recent validation of this combined approach for the bispectrum, as well as many analyses demonstrating cosmological information beyond the power spectrum, the extension to higher-order statistics has signficant potential. As the displacement-field emulator gives a model for the full tracer field, analyses involving non-standard statistics or even the full field become possible. *ksf:* my attempt at a bigger picture paragraph based on our discussion - please feel free to edit! does it belong in the intro vs discussion vs conclusions? *dsj:* I like this bigger paragraph and I think it belongs here.

In this paper, we construct a suite of three-dimensional mock galaxy fields using our forward model based on the hybrid Lagrangian bias expansion with emulated displacements from the full-field `map2map` emulator. The suite, called `muchísimocks`, consists of both real- and redshift-space mocks at $z = 0$ and spans galaxy bias and cosmological parameter space. This suite opens the door to simulation-based inference for cosmology with a hybrid bias expansion-based galaxy field, as well as with a very large number of simulations given the speed of the emulator. We demonstrate this approach by training a neural posterior inference model on summary statistics computed on `muchísimocks` in real space; in this work we focus on the galaxy–galaxy power spectrum, the galaxy–matter cross power spectrum, and the galaxy bispectrum. Our maximal training set consists of 10,000 cosmologies and 32 bias parameter sets per cosmology, for a total of 320,000 training mocks; we investigate the dependence of the inference performance on the number of training mocks, with an eye towards guiding the development of future SBI pipelines.

This paper is organized as follows. In §2, we present our forward model for galaxy fields, including details of the N-body emulator, hybrid bias expansion, and noise model. We introduce the methodology used for our simulation-based inference pipeline in §3, including the summary statistics analyzed and the construction of our training set and various test sets. In §4 we show our results on these test sets, as well as analyses of the scale dependence and convergence with size of training data. We discuss the implications of our results in §5, and summarize our findings and conclude in §6.

2 THE MUCHÍSIMOCKS SUITE

mpi: I am not sure what we decided about the RSD fields in the end.

The `muchísimocks` forward model is based on emulated N-body displacements (§2.1) and the hybrid Lagrangian bias expansion (§2.2), as well as a full-field noise model (§2.3). An overview of the model is given in the upper panel of Figure 1.

2.1 Emulating the displacement field with `map2map`

The nonlinear matter distribution can be simulated to high accuracy with N-body simulations. However, simulations with the resolution and volume required to analyze current galaxy surveys are very expensive, and training simulation-based inference models requires a very large number of simulations. We thus turn to a field-level emulator of the matter distribution, `map2map` (Jamieson et al. 2023).[1]

The density field can be approximated at linear order by the Zel'dovich approximation (ZA, also known as first-order Lagrangian perturbation theory, 1LPT), based on a realization of an initial conditions field. The `map2map` emulator learns a mapping from the ZA field at $z = 0$ to the nonlinear N-body displacement field at that redshift, effectively learning the correction from linear to nonlinear displacements. The emulator is trained on the 2000 simulations of the Quijote simulation suite that span a Latin hypercube (LH) of five cosmological parameters: Ω_c, σ_8, h, Ω_b, and n_s (Villaescusa-Navarro et al. 2020). Each has a volume of $(1\ h^{-1}\mathrm{Gpc})^3$ with 512^3 particles, giving particle mass resolutions in the range 2×10^{11} to $1\times10^{12}h^{-1}M_\odot$. Once trained, `map2map` can generate the displacement fields for any input ZA field, taking 1-2 minutes on a GPU for a $(1\ h^{-1}\mathrm{Gpc})^3$ volume *dsj:* This seems slow? What GPU are you using? On A100/H100, I get 128^3 in 0.5 s, so that should be 30 s per 1 Gpc/h box. If you run displacements and velocities sequentially on one GPU, it should be 1 min?) *ksf:* hmmm, i checked and am getting 50-100s for just the displacments; they ran on a variety of GPUs, A100s and others, and i dont see a pattern. not sure what's going on... you got 30s with your original emulator version?. [2]

In more detail, `map2map` consists of a convolutional neural network (CNN) that takes as input the ZA displacement field and outputs a prediction for the N-body displacement field; these are structured on a 3D grid as they are functions of the initial Lagrangian space field and thus can be processed as 3-channel images. As most of the cosmological parameters only affect the input power spectrum that is used to create the ZA field, these are not needed as additional inputs to the model. However, the matter density Ω_m affects the N-body evolution, so is included as an input "style" parameter; this is incorporated via the approach of the StyleGAN2 (Karras et al. 2020), in which the style parameters modulate the convolution kernels. The model uses a V-net-like architecture, with 3 layers of downsampling and upsampling, and leaky ReLU activations. For its loss function, it combines the log mean squared error of the Lagrangian particle displacements and Eulerian density field. *dsj:* could we add hyperlinks to the GitHub repo for the original pytorch implementation and a new JAX implementation? *ksf:* added; also did some rearranging so the new one is mentioned in the discussion!

For our mock suite, we define a Latin hypercube of 10,000 sets of cosmological parameters: Ω_c, σ_8, h, Ω_b, and n_s. We use uniform priors for each parameter that span $\sim10\sigma$ around the Planck Collaboration et al. (2014) values of the cosmological parameters, following Angulo et al. (2021) (with the range for σ_8 extended to slightly lower

[1] https://github.com/dsjamieson/map2map_emu

[2] We note that `map2map` can also emulate N-body velocity fields; we generate and release these as part the `muchísimocks` suite, and construct redshift-space bias fields with them, but do not use them in this paper.

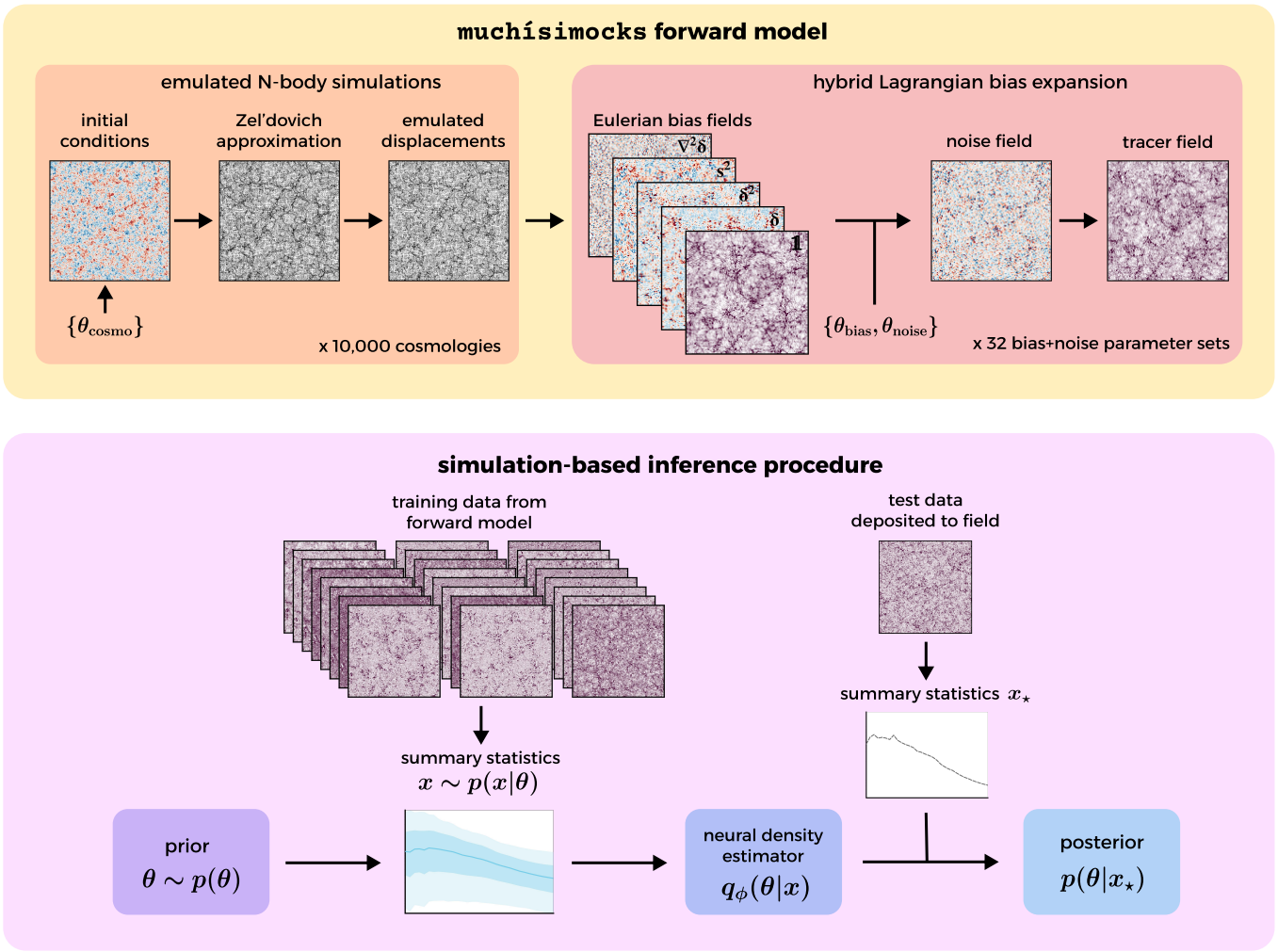


Figure 1. Schematic overview of the `muchísimocks` suite and simulation-based inference procedure. For our forward model, we start with an initial conditions field based on input cosmological parameters, and apply the Zel'dovich approximation to obtain first-order displacements; these are then are passed to the `map2map` emulator which predicts the N-body displacement field. Using these as inputs to the hybrid Lagrangian bias expansion, we produce second-order Lagrangian bias fields advected to Eulerian space; these compose the core of the `muchísimocks` suite. These are then combined with a choice of bias parameters, as well as a full-field noise model. The combined tracer fields and associated parameters are used to train a neural posterior estimator, which conditioned on test data can then predict the associated parameters. *dsj:* why "Eulerian bias field" and not "Lagrangian bias fields"? *ksf:* they are lagrangian bias fields advected to eulerian space, which does make them eulerian bias fields but emphasizes that we use the Lagrangian bias expansion, rather than directly in Eulerian space, which is different, in my understanding. does that sound right / does the language make sense to everyone? *mpi:* I love this figure.

values); these are shown in Table 1. For each of these cosmologies, we generate a ZA field based on a different random seed, and pass that field through `map2map` to obtain the nonlinear particle displacements. The `map2map`-emulated particle displacements are a key ingredient in modeling the galaxy bias fields, as discussed in the next section.

2.2 Modeling tracers with the hybrid bias expansion

To model biased tracers of the matter field, we use the hybrid Lagrangian bias expansion, also known as hybrid effective field theory (HEFT, Modi et al. 2020; Kokron et al. 2021; Zennaro et al. 2023). We first define a mapping from initial (Lagrangian) space coordinates $\boldsymbol{q}$ to Eulerian space coordinates $\boldsymbol{x}$ as $\boldsymbol{x} = \boldsymbol{q} + \Psi(\boldsymbol{q})$, where $\Psi(\boldsymbol{q})$ is the Lagrangian space displacement field. This advection from Lagrangian to Eulerian space effectively expresses the overdensity field as weighted contributions of the initial density field that end up in a given Eulerian location. *mpi:* hmmmm, I am not sure I understand what you mean here. The advection moves particles from Lagrangian to Eulerian space carrying a label. This label is the value some local functions of the overdensity field take in the Lagrangian space. Once we ask, what's the tracer overdensity field, then these labels become weights of the particle-to-mesh transformation. I would just remove the previous sentence. The displacement term $\Psi(\boldsymbol{q})$ can be computed with Lagrangian perturbation theory, but this is only accurate down to mildly nonlinear scales; here we take the "hybrid" approach which uses N-body displacements to compute $\Psi(\boldsymbol{q})$, significantly improving the accuracy down to small scales.

The galaxy overdensity field in Lagrangian space can be described to second order as

$$\begin{aligned}\delta_g(\boldsymbol{q}) = b_1\delta_L(\boldsymbol{q}) + b_2\left[\delta_L^2(\boldsymbol{q}) - \langle\delta_L^2\rangle\right] \\ + b_{s^2}\left[s^2(\boldsymbol{q}) - \langle s^2\rangle\right] + b_\Delta\nabla^2\delta_L(\boldsymbol{q})\end{aligned} \tag{1}$$

where $\delta_L(\boldsymbol{q})$ is the linear matter density in Lagrangian space; $s^2(\boldsymbol{q})$ is the shear field, defined by the traceless part of the tidal field, given by $s^2(\boldsymbol{q}) = s_{ij}(\boldsymbol{q})s^{ij}(\boldsymbol{q}) = \delta_i\delta_j\phi(\boldsymbol{q}) - \frac{1}{3}\delta^K_{ij}\delta_L(\boldsymbol{q})$, *dsj:* the LHS matches equals the middle, but they do not equal the RHS, did you mean to have two inline equation, with the second having LHD s_{ij} = ...? where $\phi(\boldsymbol{q})$ is the linear gravitational potential in Lagrangian space $\nabla^2\phi(\boldsymbol{q}) = 4\pi G\bar{\rho}\delta_L(\boldsymbol{q})$, and δ^K is the Kronecker delta. The coefficients b_1, b_2, b_{s^2}, and b_Δ are the Lagrangian bias parameters, which can be thought of as weights corresponding to each of the operators of the density field and are specific to the tracer sample being modeled. Note that the particles are advected to Eulerian space with weights $1 + \delta_g(\boldsymbol{q})$, where the 1 ensures that unbiased tracers recover the matter density field *dsj:* What does Lagrangian weighting function mean here, the Jacobian determinant? *ksf:* edited to make clearer, i mean the weights used for the particles before CIC deposition! (there's a subtlety/convention when defining $\delta_g(\boldsymbol{q})$, so this sentence just makes that clear).

We construct our tracer density fields based on this model as follows. We start from a random realization of the initial Gaussian field on a $N_{\rm grid} = 512^3$ grid in a $L = (1h^{-1}{\rm Gpc})^3$ box, matching the resolution and size of the Quijote simulations that `map2map` was trained on, and use the ZA to compute approximate displacements. We pass the ZA displacement field to `map2map` (§2.1), which outputs the emulated N-body displacement field. We compute the Lagrangian operators based on the linear density field, smoothing these with a damping scale of $k = 0.75h{\rm Mpc}^{-1}$. *ksf:* TODO test and discuss aliasing We then perform the advection to Eulerian space: The tracers are deposited onto a grid in Eulerian space with a cloud-in-cell (CIC) assignment, with positions given by the simulation displacement field and weights given by Lagrangian operators *dsj:* We smoothing then advect. Have we checked the opposite order, advect then smooth? Or is there a reason to do the smoothing in Lagrangian, not Eulerian space? *ksf:* im not sure about this, i dont think i checked explicitly; marcos/raul?*mpi:* You don't have that choice, you always need to smooth the lagrangian weights since you need to define the value of the density field your particle have in lagrangian coordinates. Therefore, the smaller smoothing scale you can get is the cell size. You can smooth again in Eulerian space, but what for? *dsj:* Actually, it is kind of obvious you should/must smooth in Lagrangian space, but we also smooth in Eularian space, that's what the $k_{\rm max}$ cut we impose amounts to. This gives five bias fields in Eulerian space, $\{O\}$: weighting the Lagrangian density field by the constant $\mathbb{1}$ gives the homogeneous field; by $\delta_L(\boldsymbol{q})$, the linear density field; by $\delta^2_L(\boldsymbol{q}) - \langle\delta^2\rangle$, the quadratic density field; by $s^2(\boldsymbol{q}) - \langle s^2\rangle$, the tidal field; and by $\Delta^2\delta(\boldsymbol{q})$ *mpi:* notation, the Laplacian. These fields can be weighted by the bias coefficients and summed to produce an overall tracer field δ_g.

We perform two postprocessing steps on these fields. To reduce storage space, we reduce the size of these fields by cutting the high-k modes: we remove modes beyond $k_{\rm Nyquist} = \pi N_{\rm grid,cut}/L \approx 0.4$ *mpi:* units, where $N_{\rm grid,cut} = 128$, our reduced grid size. We do this by Fourier transforming each bias field, nulling the modes higher than $k_{\rm Nyquist}$, and inverse Fourier transforming back to real space. At scales higher than this our hybrid bias model loses accuracy anyway and we don't consider them in our inference *mpi:* it might not be the model that loses accuracy, it could be the emulator, in my paper, kmax=0.6 h/Mpc looked good. *dsj:* how dare you blame the emulator! ;). But for real, the 2nd order bias expansion becomes insufficient at some scale. Kate, are you saying that you checked $k = 0.5h$/Mpc or $k = 0.6h$/Mpc and found degraded accuracy? Or are you making a less specific claim that at some scale with $k > 0.4h$/Mpc you know the bias expansion will eventually become inaccurate but didn't check exactly where?, so we do not give up accuracy with this k-cut. We finally perform a deconvolution of the bias fields, using a linear interpolation kernel at the original grid size $N_{\rm grid}$ to correct for the smoothing by the CIC kernel. (We confirm that this is effectively equivalent to deconvolving and then cutting the small-k modes, while being much faster.) This deconvolution could be done at the summary statistics level, but we choose to deconvolve at the field level for ease of use.

The final main suite, designed as a training set for cosmological inference, contains 10,000 sets of bias fields (5 Lagrangian bias fields advected to Eulerian space), each at a different cosmology; each has volume $(1h^{-1}{\rm Gpc})^3$ and grid size 128^3. We also generate multiple smaller suites for testing, detailed in §3.5. We call these suites `muchísimocks` (a portmanteau of the words "muchísimos", Spanish for "many", and "mocks").

2.3 Modeling noise at the field level

ksf: technically this is a postprocessing step only needed for the SBI part, is it fine here?

We require a model for stochastic noise in order to model realistic discrete galaxy populations. This can be modeled at the level of each summary statistic, but our goal is to make this suite usable for a wide range of statistics as well as field-level analyses; we thus desire a field-level noise model. Further, as we are considering higher-order statistics, we need to include non-Gaussian noise contributions. We thus turn to the noise model of Rubira & Schmidt (2025), an EFT approach described by nonlinear couplings of a single Gaussian field to the bias fields. In this model the galaxy density field can be written as

$$\delta_g(\mathbf{x}) = \sum_{m=0}^{\infty}\sum_{\mathbb{1},O} b_O^{\{m\}}\,[\epsilon_G(\mathbf{x})]^m\,O(\mathbf{x}) \tag{2}$$

where m is the order of the expansion in the noise field, $\epsilon_G(\mathbf{x}) = \mathcal{N}(O,1)$ is a unit Gaussian field, O are operators of the galaxy density field, and $\{b_{\mathbb{1}}^{\{m\}}, b_O^{\{m\}}\}$ are free coefficients. In our case we limit to second order in the bias expansion, as described in §2.2. Limiting to zeroth order in m reduces to the noiseless case, as in Equation 1. For our SBI analysis, we include terms through first order in the noise expansion, $m = 1$. That gives us bias coefficients $\{b_O^{\{0\}}\} = b_1$, b_2, b_{s^2}, b_Δ, as in Equation 1 (we fix $b_{\mathbb{1}}^{\{0\}} = 1$), and noise coefficients $\{b_O^{\{1\}}\} = A^{\rm noise}_{\rm homog}, A^{\rm noise}_{b_1}, A^{\rm noise}_{b_2}, A^{\rm noise}_{b_{s^2}}, A^{\rm noise}_{b_\Delta}$.

We note that limiting the noise model to $m = 1$ means that we exclude noise contributions to $P_{gm}(k)$, as at $m = 1$ the noise field ϵ_G has zero mean (whereas $P_{gg}(k)$ and $B_{ggg}(k_1,k_2,k_3)$ get factors of ϵ_G^2 which don't vanish). This means that the bias parameters may have to absorb partial noise contributions, biasing the inference of the bias parameters themselves; future work could extend this noise model to $m = 2$ or choose a well-motivated subset of noise parameters tailored to the statistics analyzed.

Operationally, we generate one unit Gaussian noise field ϵ_G per cosmology in our suite. We store these and use them to construct the noisy tracer field on-the-fly just before computing the summary statistics of the field. This means that we use the same noise realization when we select different bias and noise parameter sets for the same cosmology (as detailed in §3.2 and §3.5), which introduces some undesirable non-independence; however, this allows for easy reproducibility, and we expect the effect to be minimal, while the storage cost of one noise field per bias model is prohibitive. *dsj:*

could we generate more noise fields for free without storing anything additional by simply shuffling the $\epsilon_G x$ field values? Also, isn't generating the noise fields cheap? Why not generate new one one-the-fly rather than save and reusing one? For reproducibility, you could just log random seeds used to generate the noise fields.

3 SIMULATION-BASED INFERENCE METHODOLOGY

Our SBI approach uses neural posterior estimation (§3.1) trained on the main `muchísimocks` Latin hypercube (§3.2) with summary statistics computed on the mocks (§3.3). We determine implementation and training details (§3.4) and apply our final pipeline to a range of test sets (§3.5). An overview of the SBI training and testing procedure is shown in the lower panel of Figure 1.

3.1 Neural posterior estimation

The main idea behind simulation-based inference is that we can infer a posterior distribution of our data over the parameters of interest using only a *simulator*, which produces data samples given a set of parameters. This is distinct from explicit likelihood inference, in which we are required to define a likelihood of our data with respect to the parameters. Neural density estimators (NDEs) present a flexible and robust way to model a target distribution *mpi:* citation here maybe?. Here, we use neural posterior estimation (NPE), in which the posterior distribution is directly approximated with an NDE.

To do this, we aim to minimize the distance between the true posterior $p(\theta|x)$ and the NDE model of the posterior $q_\phi(\theta|x)$, where ϕ are the weights of the neural network model. We use maximum likelihood estimation, which minimizes the Kullback-Leibler divergence between the distributions. In practice, we consider the posteriors multiplied by $p(x)$, and use the fact that $p(\theta|x)p(x) = p(\theta, x)$:

$$\begin{aligned}\min_\phi D_{\mathrm{KL}}\big(p(\theta|x)\,\|\,q_\phi(\theta|x)\,p(x)\big) &= \min_\phi \int p(\theta,x)\,\log\frac{p(\theta|x)}{q_\phi(\theta|x)}\,d\theta\,dx \\ &\approx \min_\phi \sum_i \log p(\theta_i|x_i) - \log q_\phi(\theta_i|x_i) \\ &\approx \min_\phi \sum_i -\log q_\phi(\theta_i|x_i) \\ &\approx \max_\phi \sum_i \log q_\phi(\theta_i|x_i)\end{aligned} \tag{3}$$

where in the second line, we perform a Monte Carlo expectation of the integral using discrete samples from our simulation (dropping the $\frac{1}{N}$ as it does not affect the minimization); in the third line, we observe that we are minimizing with respect to the neural network weights ϕ and the true posterior $p(\theta|x)$ is independent of ϕ so it drops out; and in the final line we use the fact that minimizing the negative is the same as maximizing the positive.

Normalizing flows are a popular choice for the NDE, thanks to their expressivity and flexibility. The key idea is to transform an initial simple distribution into the target distribution through a chain of transformations. A normalizing flow is defined as

$$q_\phi(\theta|x) = \pi_z\big(f^{-1}(\theta,x)\big)\left|\det\left(\frac{\partial f^{-1}}{\partial\theta}\right)\right|, \tag{4}$$

where f is an invertible transformation modeled as a neural network and π_z is a base distribution, for which we follow the common choice of a Gaussian, $\pi_z(z) = \mathcal{N}(z|\mathbf{0},\mathbf{I})$. The Jacobian (the determinant term) arises from the change of variables, and involves f^{-1}, thus requiring f to be invertible. The invertibility property of NFs both enables us to sample from the network and gives us an explicit density, which is key for performing SBI.

Specifically, we use masked autoregressive flows (MAFs), which stack Masked Autoencoders for Distribution Estimation (MADEs) in order to model more complex distributions:

$$q_\phi(\theta \mid x) = \pi_z(z_0)\prod_{t=1}^{T}\left|\det\left(\frac{\partial f_t^{-1}}{\partial z_t}\right)\right|, \tag{5}$$

where $z_0 \sim \pi_z$ and t indexes the MADE blocks. Each network f_t takes as input the previous variable z_{t-1} as well as the observation x, and the final layer is identified with the parameters of interest, $z_T \equiv \theta$. The loss function derived from the KL divergence (Equation 3) is used to train the network and learn the optimal weights and biases.

3.2 Training set

To train our simulation-based inference model, we use the main `muchísimocks` suite of second-order bias fields to construct a set of tracer fields. Our prior on the cosmological parameters is set by the range of the suite, as described in 2.1. We need to define the range of the bias coefficients $\{b\}$ we use to sum *mpi:* weight? the fields; for this, we turn to the results of Zennaro et al. (2022), which define priors that capture a range of realistic galaxy populations modeled with the extended subhalo abundance matching technique (Contreras et al. 2021). We choose to use uniform priors with slightly wider ranges than the relations found in this work, a more conservative choice; these prior ranges are shown in Table 1. For the noise coefficients $\{A^{\mathrm{noise}}\}$ we choose wide uniform priors loosely based off of the corresponding (noiseless) bias coefficients, with empirical checks ensuring that these vary the statistics within a reasonable range *dsj:* this is a bit vague, how do we define varying within a reasonable range?. The prior ranges of the bias and noise parameters used in our SBI analysis are shown in Table 1.

While we are limited to the 10,000 cosmologies defined in the `muchísimocks` suite, we can combine these into tracer fields with any number of bias parameter sets we choose. At a given cosmology, we then have multiple tracer fields generated with distinct sets of bias and noise parameters. Though this means the final tracer fields are not fully independent of each other, this has been shown (e.g. in Hahn et al. (2023c)) to improve inference by significantly augmenting the training data size and allowing the model to disentangle cosmology and bias parameters. Note that the summing of bias fields with the chosen bias coefficients is essentially instantaneous and can be done in postprocessing just before computing summary statistics, and thus requires no additional compute time or storage (though we are still limited by the time required to compute summary statistics on these fields).

As we later want to test convergence of the inference with the number of bias parameter sets, we use a *nested Latin hypercube* to generate the bias parameters. The top-level LH contains 320,000 points, resulting in 32 independent bias parameter sets for each of the 10,000 cosmologies. Nested subsets of this LH are also LHs, with 16, 8, 4, 2, and 1 bias parameter sets per cosmology. We generate a nested LH for the 4 bias parameters and 5 noise parameters, for a total of 9 independently varied parameters.

This was achieved with the Qian (2009) algorithm for nested Latin hypercubes, which works by starting with a two-layer structure, with coarse cells and fine cells on a multidimensional grid. *dsj:* I find

Table 1. Uniform prior ranges for the cosmological, bias, and noise parameters. The cosmological parameters are based on a $\sim 10\sigma$ range around the Planck Collaboration et al. (2014) best-fit values, and the bias and noise parameters are loosely based on the hypervolume found by Zennaro et al. (2022) for a wide range of realistic galaxy populations.

Parameter	Lower bound	Upper bound
Cosmological parameters		
Ω_c	0.23	0.40
σ_8	0.65	0.90
h	0.6	0.8
Ω_b	0.04	0.06
n_s	0.92	1.01
Bias parameters		
b_1	−1.0	3.0
b_2	−2.0	2.0
b_{s^2}	−2.0	2.0
b_Δ	−10.0	10.0
Noise parameters		
$A^{\rm noise}_{\rm homog}$	−3.0	3.0
$A^{\rm noise}_{b_1}$	−3.0	3.0
$A^{\rm noise}_{b_2}$	−2.0	2.0
$A^{\rm noise}_{b_{s^2}}$	−5.0	5.0
$A^{\rm noise}_{b_\Delta}$	−10.0	10.0

the paragraph break and then opening with "This" a bit awkward. It constructs a standard LH on the coarse cells, and then fills in the finer grid, also ensuring it creates a valid LH. It then performs random permutations of rows and cells that still preserve the LH structure but remove any deterministic structure from the filling procedure, resulting in a random realization of two nested LHs. This is then repeated for subsequent layers by adding finer and finer grids. A python implementation of this algorithm along with validation tests are available in the `gethypercube`[3] package. *ksf:* i may include an appendix detailing my nested-LH package! thoughts?*mpi:* anything is fine with me *dsj:* Me too, go for it! Though the LHs at all levels are exact (the samples live in disjoint intervals across all the dimensions), they are not guaranteed to have optimal space-filling properties compared to an optimized single-level LH. Optimizations are possible but very time-consuming, so we use the standard nested hypercube (with the randomization described above); this still achieves LHs that have significantly better space-filling properties across all levels than either taking random subsets of a top-level LH, or stacking bottom-level LHs. The use of LH-distributed parameters has been shown to be important for accurate and robust training, so we expect that this nested strategy leads to robustness in our convergence tests, which are a key result of this work *mpi:* it is unclear to me if we achieved this in the end because of the later issues with convergence, maybe we can soften these claims? Otherwise we didn't achieve a key result of this paper. *dsj:* I would think scrambled Sobol sequence would be more optimal and naturally extensible, so the nesting thing isn't an issue for adding bias/noise samples on top of cosmology. If provides uniform coverage not just for the whole parameter set, but also for projected subspaces for parameters, which LHC doesn't.

To generate our final tracer fields, we weight the `muchísimocks` bias fields and noise field by the selected set of corresponding bias and noise parameters as in Equation (2), and sum these weighted fields. We normalize these fields by the original `map2map` grid size (512^3) before computing their summary statistics.

[3] http://github.com/kstoreyf/gethypercube

3.3 Summary statistics

We perform our SBI analysis with our `muchísimocks` framework on three standard summary statistics: the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter power spectrum $P_{gm}(k)$, and the bispectrum $B_{ggg}(k_1, k_2, k_3)$.

The galaxy–galaxy auto power spectrum $P_{gg}(k)$ is defined as

$$\langle \delta_g(\boldsymbol{k}) \delta_g(\boldsymbol{k'}) \rangle = (2\pi)^3 \delta_{\rm D}(\boldsymbol{k} + \boldsymbol{k'}) P_{gg}(k)\,, \quad (6)$$

where $k = |\boldsymbol{k}|$ is the Fourier space wavenumber. This is the two-point function of the galaxy field, the most common statistic for cosmological inference from galaxy samples, which captures all the Gaussian information. We use 31 logarithmically spaced bins from $k = 0.01$ to $k \simeq 0.37$; the maximum k value is set to be below the Nyquist frequency of our mocks. We use the BACCO code to compute $P_{gg}(k)$ for our mocks. We can compute the 15 cross-terms of the 5 bias fields and later sum their contributions based on the bias parameters to obtain the power spectrum of the total tracer field; in practice we do this for the noiseless case, but in the noisy case compute $P_{gg}(k)$ for the total tracer field as this would add additional terms.

The galaxy–matter cross power spectrum $P_{gm}(k)$ is defined by

$$\langle \delta_g(\boldsymbol{k}) \delta_m(\boldsymbol{k'}) \rangle = (2\pi)^3 \delta_{\rm D}(\boldsymbol{k} + \boldsymbol{k'}) P_{gm}(k)\,, \quad (7)$$

where $\delta_m(\boldsymbol{k})$ is the matter density field in Fourier space. While not directly observable as it relies on the matter field, this represents the galaxy-galaxy lensing contribution to 3x2pt analyses. We use the same binning as $P_{gg}(k)$, and again use the BACCO code to compute $P_{gm}(k)$ for our mocks; for our fiducial results, we only use up to $k_{\rm max} = 0.25$ for $P_{gm}(k)$.

The galaxy auto-bispectrum $B_{ggg}(k_1, k_2, k_3)$ is defined by

$$\langle \delta_g(\boldsymbol{k}_1) \delta_g(\boldsymbol{k}_2) \delta_g(\boldsymbol{k}_3) \rangle = (2\pi)^3 \delta_{\rm D}(\boldsymbol{k}_1 + \boldsymbol{k}_2 + \boldsymbol{k}_3) B_{ggg}(k_1, k_2, k_3)\,. \quad (8)$$

This is the Fourier space three-point function. We choose 12 linearly spaced bin edges from $k = 0.01$ to $k = 0.4$, yielding 203 triangles; for our fiducial results, we only use triangles with edges up to $k_{\rm max} = 0.25$, yielding 50 triangles. *dsj:* why log spaced bins for power spectrum but linear spaced bins for bispectrum? We use the `PolyBin3D` code (Philcox & Flöss 2024) to compute $B_{ggg}(k_1, k_2, k_3)$ for our mocks. *mpi:* should we include here on which papers did we base these scale cuts on? I might be wrong but I don't remember stating clearly that these are based on my previous works, which is completely fine. Also, maybe we could make clearer that this work is a natural continuation of the field level model we did with Drew? I'm just thinking aloud.

The summary statistics computed on the `muchísimocks` training set are shown in Fig. 2.

3.4 Implementation and training

Before training, we reparameterize the cosmology and bias parameters to combinations that are better constrained by the summary statistics, and thus should result in improved training convergence. We multiply the bias and noise parameters by the relevant power of σ_8, obtaining $\sigma_8 b_1$, $\sigma_8^2 b_2$, $\sigma_8^2 b_{s^2}$, $\sigma_8 b_\Delta$, $\sigma_8 A^{\rm noise}_{b_1}$, $\sigma_8^2 A^{\rm noise}_{b_2}$, $\sigma_8^2 A^{\rm noise}_{b_{s^2}}$, and $\sigma_8 A^{\rm noise}_{b_\Delta}$. When sampling from the posterior, we can divide out the appropriate power of the recovered σ_8 to obtain estimates of the original bias and noise parameters. We also do not

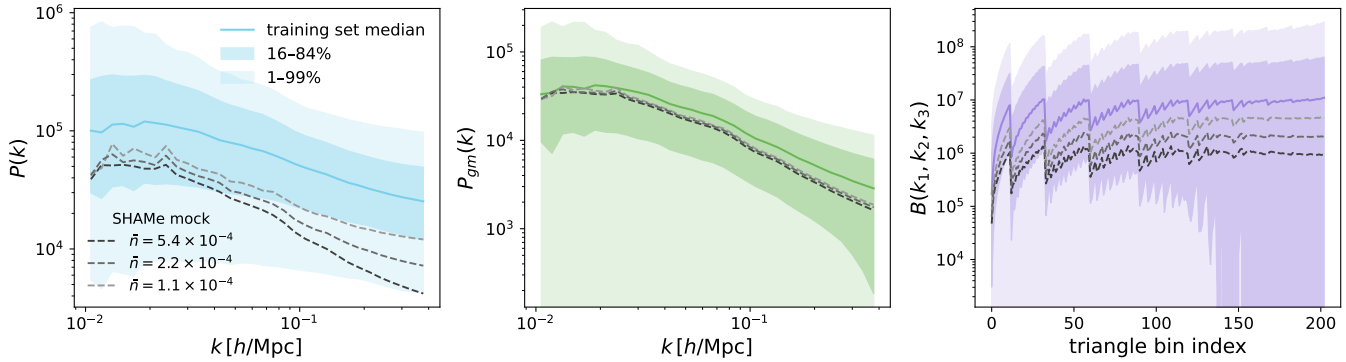


Figure 2. The summary statistics computed on the `muchísimocks` training set: the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–mater power spectrum $P_{gm}(k)$, and the galaxy bispectrum $B_{ggg}(k_1, k_2, k_3)$. The median, 16th-84th percentiles, and 1st-99th percentiles of the training set are shown. Out-of-distribution test mock catalogs constructed with extended subhalo abundance matching (SHAMe) on a BACCO simulation is also shown, for three number densities (dashed). In our fiducial inference model, scales below $k_{\max} = 0.25$ for $P_{gm}(k)$ and triangles with sides smaller than $k_{\max} = 0.25$ for $B_{ggg}(k_1, k_2, k_3)$ are excluded.

include the cosmological parameters h, Ω_b, and n_s in the SBI training, as we do not expect to constrain these parameters (and as confirmed in tests); these are still varied in the training set and implicitly marginalized over.

We use the publicly available `sbi` (Tejero-Cantero et al. 2020) package for the NPE and MAF implementations (Papamakarios & Murray 2016; Greenberg et al. 2019); the MAF uses the `nflows` Python package (Durkan et al. 2019, 2020). Our data vector is comprised of the binned statistics discussed in §3.3, concanenated when using multiple statics *mpi:* statistics?. The statistics combinations we show are: $P_{gg}(k)$, $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$, $P_{gg}(k) + P_{gm}(k)$, $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ (we exclude $B_{ggg}(k_1, k_2, k_3)$ and $P_{gm}(k)$ alone, as these do not contain much constraining power on their own and in practice are almost always considered in combination with $P_{gg}(k)$). We choose a 90%/10% training/validation split of our training set. We use the Adam optimizer, and stop training when the mean loss over the validation set does not improve for 20 epochs; we save the network weights at the best validation loss.

We perform a sweep over several hyperparameters, using 30 random samples from the following sets: the learning rate $\{10^{-2}, 3\times10^{-3}, 10^{-3}, 3\times10^{-4}, 10^{-4}, 3\times10^{-5}, 10^{-5}\}$, hidden features {32, 64, 128, 256}, training batch size {32, 64, 128, 256}, and number of transforms {4, 6, 8, 10}. The sweeps are performed separately for each statistic or combination of statistics, and for each choice of noiseless (bias parameters only) and noisy (bias and noise parameters) training data; we use the maximum size training set of 320,000 data vectors, and for the statistics use $k_{\max}^{P_{gg}} = 0.37$, $k_{\max}^{P_{gm}} = 0.25$, and $k_{\max}^{B} = 0.25$. We look at the ordered mean validation losses of the 30 trained models for each statistics combination, and observe a signficant drop for the top three models (this number will vary depending on the number models explored in the sweep and is generally chosen empirically). We thus choose to use an equally weighted ensemble of the top three MAFs when performing inference, which has been shown to lead to more stable results (e.g. Alvey et al. 2026); explicitly, we sample the MAFs as $q_\phi(\theta \mid x) = \frac{1}{3}\sum_{j=1}^{3} q_\phi^j(\theta \mid x)$. We use the sets of hyperparameters from the three top MAFs to train variations of the model for the given statistics combination, such as when varying the number of training points and the $k_{\max}$ values, rather than running dedicated sweeps which would be computationally very expensive. The set of best identified hyperparameters vary across the statistics, with the noisy case (which varies more parameters) generally preferring larger hidden feature vectors and batch sizes, though most learning rates in both noisy and noiseless cases are around 10^{-4}.

Training one MAF on the maximum size training set of 320,000 data vectors takes a few hours on a single CPU for the fiducial model described above. To display the posteriors and compute covariances and other statistics from them, we draw 10,000 samples from the posterior; we confirm that the posteriors are generally converged with this number of samples.

Table 2. Overview of the `muchísimocks` mock suites, as well as additional out-of-distribution test mocks. The label `mm` indicates the `muchísimocks` forward model; LH is a Latin hypercube; Planck is the Planck Collaboration et al. (2014) cosmology. For the LHs, the prior ranges for the parameters are those in Table 1. N_{bias} indicates the total number of bias and noise parameter sets; the training set uses 32 per cosmology, while the coverage and cosmic variance sets use 1 per cosmology.

Suite	$N_{\text{realizations}}$	$N_{\text{cosmologies}}$	N_{bias}
`mm` training set	10,000 (LH)	10,000 (LH)	320,000 (LH)
`mm` coverage test set	1,000	1,000 (LH)	1,000 (LH)
`mm` cosmic variance test set	1,000	1 (Planck)	1 (matched to SHAMe $\bar{n} = 2.2\times10^{-4}$)
BACCO + SHAMe mock	1	1 (Planck)	3 ($\bar{n} = \{1.1, 2.2, 5.4\}$ $\times10^{-4}$)

3.5 Test sets

We generate several test sets to validate our `muchísimocks` suite and our SBI approach. An overview of these test suites, as well as the training set, is shown in Table 2.

Our first test set is a *cosmic variance set*. For this we use the `muchísimocks` pipeline to construct 1,000 additional mocks each with a different random seeds *mpi:* seed? *dsj:* nope, seeds is correct, all at the same fixed cosmology; we choose this to be the same cosmology as the out-of-distribution SHAMe mock test described below: $\Omega_c = 0.309$, $\sigma_8 = 0.816$, $h = 0.677$, $\Omega_b = 0.0486$, and $n_s = 0.967$. We fix the bias parameters to those measured from that

mock, and use these to combine the bias fields into a tracer field for each realization. For the noise parameters, we cannot measure these from the mock directly (they are very poorly constrained); to identify realistic noise parameters, we generate 1,000 sets of noise parameters and add the resulting noise field to each of the 1,000 mocks, obtaining tracer fields that have the same cosmology and bias parameters but different noise parameters. We compute the statistics of these fields, and then compute the reduced χ^2 between them and the statistics of the SHAMe mock with fiducial number density $\bar{n} = 2.2 \times 10^{-4}\,(h^{-1}\mathrm{Mpc})^{-3}$. Specifically, we use $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ with the fiducial $k_{\rm max}$ choices (the noise model does not affect $P_{gm}(k)$, as discussed in §2.3, so we exclude it), and choose the set of noise parameters that gives the lowest χ^2. We then use this single set of SHAMe-mock-matched noise parameters to compute the noise terms of our model, using 1,000 different Gaussian noise field realizations ϵ_G, and combine this with the noiseless term from only the bias fields to obtain our final set of 1,000 cosmic variance mocks at fixed cosmology, bias, and noise parameters, only varying the random seeds.

Our second test set is our *coverage set*. This consists of 1,000 mocks generated with the `muchísimocks` pipeline that span the same prior range of cosmology, bias, and noise parameters as in the training set. To select the parameters, we construct an independent LH of size 1,000 for the cosmologies, and a separate LH for the bias and noise parameters, with a single set of bias and noise parameters per cosmology for a total of 1,000 mocks. The coverage set tests the performance of our method across the prior space and its dependence on the parameter values.

Our final test set is our *SHAMe mock mpi:* maybe we can call it SHAMe high fidelity mock? To distinguish it from the other mocks?; we construct a realistic catalog of galaxies using the extended subhalo abundance matching (SHAMe) approach (Contreras et al. 2021) applied to a BACCO N-body simulation Angulo et al. (2021). This is a stringent test as it uses both an independent, signficantly higher resolution underlying cosmological simulation and a completely different galaxy formation model than the `muchísimocks` mocks used to train the inference network, testing the generalizability of our model. In detail, we use a simulation with a box size of $(1024\ h^{-1}\mathrm{Mpc})^3$ (slightly larger than the `muchísimocks` training data) and 3072^3 particles, which uses the fixed-and-paired technique (Angulo et al. 2021), and has Planck cosmology $\Omega_{\rm c} = 0.309$, $\sigma_8 = 0.816$, $h = 0.677$, $\Omega_{\rm b} = 0.0486$, and $n_{\rm s} = 0.967$ (Planck Collaboration et al. 2014). Halos are identified using a variant of the SUBFIND halo finder (Springel et al. 2001), and we populate these halos with galaxies using the SHAMe model with the following parameter choices: $\sigma_{\rm logM} = 0.47513517$, $t_{\rm merger} = -0.4577029$, $f_{\rm k,cen+sat} = 0.20147321$, $f_{\rm k,cen-sat} = -0.02964998$, $\beta_{\rm lum} = 14.82600407$. *ksf:* think it's ok to use fewer sigfigs here?*mpi:* defenitelly, and also, can you double check if none of these parameters have units? We choose a fiducial number density of $\bar{n} = 2.2 \times 10^{-4}\,(h^{-1}\mathrm{Mpc})^{-3}$, similar to that of DESI luminous red galaxies (LRGs) when scaled to $z = 0$. *ksf:* checking that this is right? also are the shame parameters chosen to be lrg-like? To check our inference results, we measure the bias parameters of this sample using the field-level probabilistic bias approach of Stücker et al. (2025b,a); we find $b_1 = 0.474$, $b_2 = 0.127$, and $b_{s^2} = -0.339$ (we do not expect b_Δ to be well-measured by this approach *mpi:* this is not true, we expect this approach to measure b_Δ very well, but it depends on the selected smoothing scale. Note that b_Δ has units and, therefore, different smoothing scales will give different values. The problem is that, we cannot select the same smoothing scale in the bias fields for the probabilistic bias, because it is too small for the peak background split approximation. Just write that we cannot predict its value independently of a given smoothing scale.). We also test a higher and lower number density sample, of $5.4 \times 10^{-4}\,(h^{-1}\mathrm{Mpc})^{-3}$ and $1.1 \times 10^{-4}\,(h^{-1}\mathrm{Mpc})^{-3}$; these are measured to have slightly different bias parameters. The summary statistics computed on the SHAMe mock are shown in Fig. 2; they largely fall within the middle 68% of the training set, with the exception of $P_{gg}(k)$ at smaller scales, which reaches the edge of the training set distribution but still falls within it.

For each of the two simulations of opposite phases from the fixed-and-paired pair, the SHAMe model gives a set of discrete galaxy tracers. We deposit these using the Cloud-in-Cell mass assignment scheme to a mesh of size 524^3 to achieve approximately the same cell size as the `muchísimocks` mocks given the slight difference in box size. We then remove the high-k modes to downsample to 132^3, again to match the `muchísimocks` cell size, and deconvolve the field in the same way as the `muchísimocks` fields, based on the original grid size. After turning these into normalized overdensity fields, we compute the statistics on each of the two fields with the same method used for the training data, and take the mean of these two statistics as the data vector for inference.

4 SBI RESULTS

4.1 In-distribution recovery tests

4.1.1 Cosmic variance test set results

We first validate our SBI pipeline on the cosmic variance test set, at fixed values of the cosmology, bias, and noise parameters. We compute summary statistics on the test data in the same way as on the training data.

For the cosmic variance test set, we compute the mean of the statistics computed on the 1,000 individual realizations, and use this as the data vector for inference. In Appendix A, we show that this gives consistent posteriors as combining the results of inference on each of the 1,000 mocks, with the mean and MAP closely agreeing; the posterior widths do differ by up to 5%, but we expect the conclusions from this validation test to hold. *dsj:* I don't understand what the second part of this sentence means, about expecting the conclusions to hold. We expect them to hold when? In the more stringent test with out-of-distribution data? They don't agree on the width of the posterior, but one is the correct thing, and the other is not, so we should state what direction the disagreement is in. It looks to me like the inference-on-mean runs are 5% overconfident, is that right? This inference on the mean has the benefit of averaging out the effects of cosmic variance to obtain the mean constraint and allowing us to visualize a typical contour, as well as being signficantly faster. *dsj:* This is only an benefit if training hasn't produced a model that learns something about the cosmic variance and performs poorly in absence of it. The posteriors are shown in Figure 3 for three key parameters, $\Omega_{\rm m}$, σ_8, and b_1; we infer and marginalize over the other bias parameters and the noise parameters. (The full contour plot is shown in Fig. B1.)

While $P_{gg}(k)$ on its own provides significant constraining power on $\Omega_{\rm m}$, it results in a weak constraint on b_1 and a weak, biased constraint on σ_8 *mpi:* just for curiosity, do we know if S_8 suffers from the same problem or would it be an alternative? *dsj:* Based on the contour in the $\Omega_{\rm cold}$ column, the principle axes are already vertical/horizontal, so unless some weird project effects are happening, $\Omega_{\rm cold}$ is already the tightest-constrained direction in that subspace. The S_8-type parameter is more relevant for the weak lensing/cluster count situation, where you measure the amplitude of density fluc-

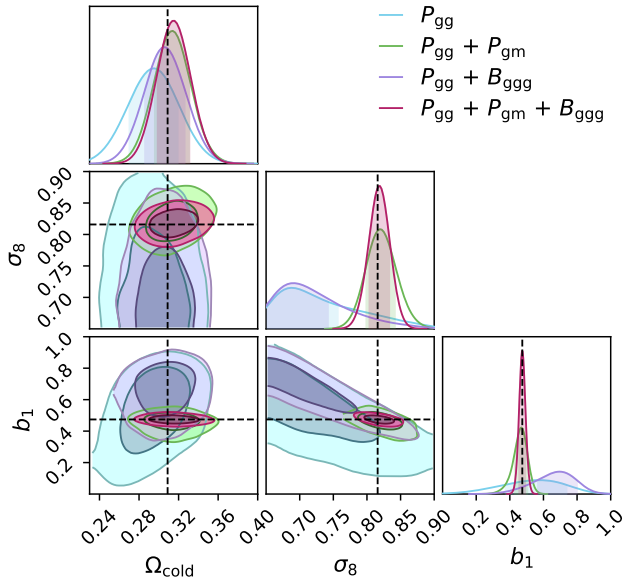


Figure 3. Posteriors for the key parameters Ω_m, σ_8, and b_1 inferred from the mean of the statistics over our fixed-parameter cosmic variance test set, for each of the statistics combinations considered. The inclusion of $P_{gm}(k)$ breaks the $\sigma_8 - b_1$ degeneracy, providing tight and unbiased constraints on the parameters; adding $B_{ggg}(k_1, k_2, k_3)$ tightens the constraints. The other bias parameters and the noise parameters are inferred and marginalized over; the full posterior plot is shown in Fig. B1.

tuations weighted by how much matter there is to lense/collapse, so Ω_m affects the amplitude itself $\sim \sigma_8\Omega_m^a$ for some a. That isn't the situation with P_{gg}, where we directly use the 3D distribution of galaxies to trace the underlying matter field.. The weakness of these constraints is expected as a result of the perfect degeneracy between σ_8 and b_1 in the power spectrum (as $P_{gg}(k) = (b_1\sigma_8)^2\hat{P}(k)$); we check that the combination $\sigma_8 b_1$ is accurately constrained. Including $B_{ggg}(k_1, k_2, k_3)$ interestingly only slightly increases the precision on Ω_m; further, it does not improve the bias on σ_8 or b_1. Notably, it does signficantly increase the precision on the higher-order bias parameters b_2 and b_{s^2}. We see that using the statistics combination $P_{gg}(k) + P_{gm}(k)$ produces unbiased constraints on σ_8 and b_1; this is expected as $P_{gm}(k)$ is able to break their degeneracy (as $P_{gm}(k) = b_1\sigma_8^2\hat{P}(k)$). This corroborates our hypothesis that the bias observed in σ_8 for $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ is likely a result of projection effects caused by a lack of constraining power. The full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ preserves the unbiased constraints, and the bispectrum adds an additional improvement of constraining power compared to $P_{gg}(k) + P_{gm}(k)$: it improves Ω_m by 6%, σ_8 by 49%, and b_1 by 2.1×; it has an even stronger effect on the bias parameters, improving b_2 by 3.5×, b_{s^2} by 2.9×, and b_Δ by 2.5×. The noise parameters are all poorly constrained, recovering the prior; however, we check that marginalizing over these is necessary and sufficient for unbiased constraints.

These tight, unbiased constraints on the cosmology and bias parameters with the full data vector considered serve as a validation of our inference approach. We note that these results are for real space, and we would expect differences in the relative information content in the statistics in redshift-space, where there is additional cosmological information contained in redshift-space distortions.

4.1.2 Coverage test set results

We infer the parameters of the 1,000 samples of our coverage test set, spanning the prior range of our training set; this allows us to test the dependence of the estimated parameters on the true parameter values. Note that the prior has a wide range of values of the noise parameters, so these test samples will have a variety of noise levels.

In Figure 4, we show the differences between the predicted values and true values of the three key parameters parameters as a function of the true value; we display the median of the 1,000 mocks, and the 16-84th percentile range. The predicted value is the mean of the inferred posterior; we check that the behavior is very similar for the maximum a posteriori (MAP) values. For $P_{gg}(k)$, we see that for all of the parameters there is a bias that increases with distance from the center of the prior, with a bias high at smaller true values and low at higher values. This is expected when the statistic does not provide significant constraining power on the parameter, as is the case with $P_{gg}(k)$, as the prediction will tend towards the mean. Including $B_{ggg}(k_1, k_2, k_3)$ improves the parameter-dependent bias for all parameters, and also generally tightens the range of the residuals. For $P_{gg}(k) + P_{gm}(k)$, we can see that the bias on σ_8 improves significantly, as $P_{gm}(k)$ breaks the degeneracy with b_1 and thus provides stronger constraining power on all parameters and avoids projection effects, as we saw in the cosmic variance test set. $P_{gm}(k)$ also improves the accuracy and precision of the constraints on Ω_m and b_1. Further including $B_{ggg}(k_1, k_2, k_3)$ helps marginally for all parameters, and this full parameter combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ yields largely unbiased results across the coverage set, indicating that our inference is robust across the prior. There are still slight biases for $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ at the edges of the prior range, particularly at high Ω_m and σ_8, so the inference should be considered to be most robust with a small buffer from the prior edges.

The residuals shown above only quantify the accuracy of the means of the posteriors; to assess the inferred uncertainties, we compute the coverage of the samples, defined as the fraction of samples for which the true value falls within the predicted uncertainty range, as a function of the given level of uncertainty. For instance, for a credibility level of 68%, 68% of the samples should have the truth fall within their predicted 68% error region ("empirical coverage"). A higher empirical coverage compared to credibility level indicates underconfidence, while lower indicates overconfidence. This coverage (or probability-probability) plot is shown in Figure 5, for the 500 mocks of the coverage set closest to the center of the prior in Ω_m, σ_8, and Ω_m *mpi: b_1?*. We see that the inferred uncertainties are generally quite well calibrated for all three key parameters. Using only $P_{gg}(k)$ or $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ results in a slight underconfidence in the posteriors, at the level of $0.1 - 0.25\sigma$ (assuming a Gaussian distribution centered on the truth). The full statistics combination $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ results in improved calibration of the uncertainties, with only slight fluctuations. We note that for this check, we have excluded the mocks on the edges of the prior to isolate the assessment of the SBI-inferred uncertainties; when we make the coverage plot for all 1,000 mocks of the coverage set, we do see some overconfidence. This is a result of the biases towards the edges of the prior which cause the uncertainty regions to not sufficiently cover the truth, and the uncertainty estimates do not fully compensate for this bias.

These results show that our inference pipeline can accurately recover the cosmological and bias parameters across a range of parameter values. *mpi:* I find this sentence a bit too vague. If you intend to make a summary, can you specify the range?

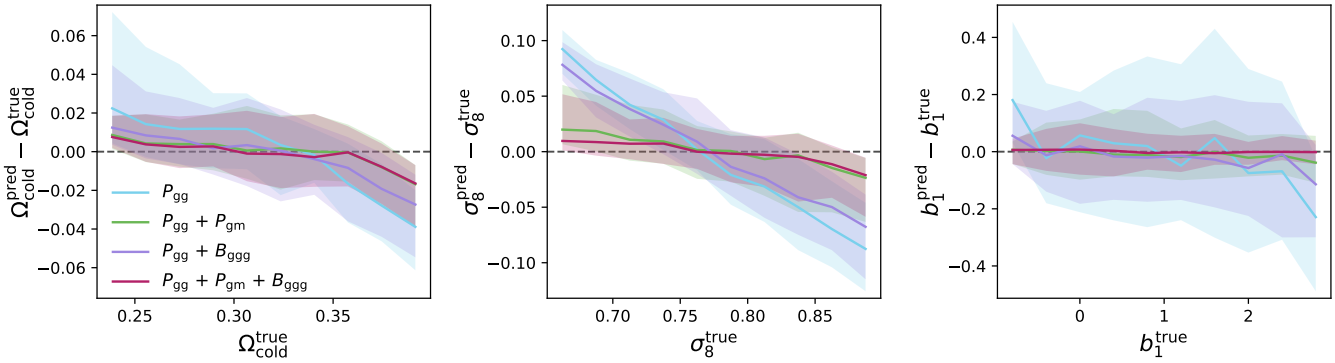


Figure 4. The difference between the predicted (mean of the posterior) and true parameter values for the key parameters for each of the 1,000 coverage test set mocks, as a function of the true value. The median and 16th-84th percentile ranges in bins of the true value are shown for each of the stastistics combinations considered. Using the full statistics combination gives largely unbiased recovery across the prior range, with a small biases at the high ends of $\Omega_{\rm m}$ and σ_8; using fewer statistics results in stronger parameter-dependent biases that tend towards the mean. Including more statistics also results in tighter parameter constraints, most notably with $P_{gm}(k)$ significantly improving the precision and bias on all key parameters.

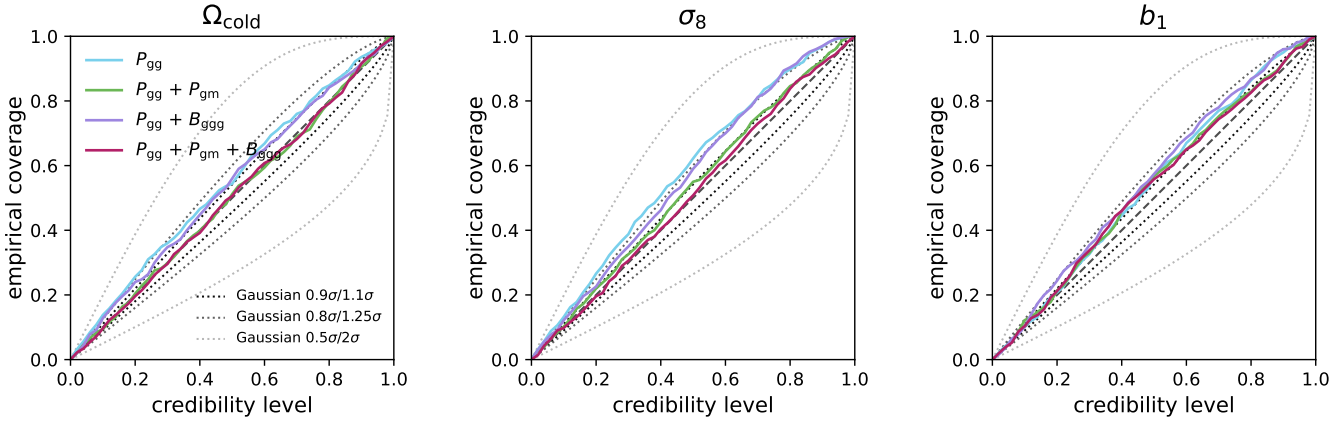


Figure 5. Coverage (probability–probability) plots for the key parameters for the 500 mocks of the coverage test set closest to the center of the prior. The empirical coverage is the fraction of test set points with predicted uncertainties at the given credibility level that encompass the true value. The closeness to the 1:1 line indicates well-calibrated uncertainties; dotted lines show the expected deviations if posterior widths were off by the given factor of σ, assuming Gaussians centered on the truth. The posteriors are largely well-calibrated, with the full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ giving the best calibration across parameters.

4.2 Out-of-distribution recovery test: SHAMe mock

The results of our SBI pipeline on our SHAMe out-of-distribution test mock with fiducial number density $\bar{n} = 2.2 \times 10^{-4}\,(h^{-1}{\rm Mpc})^{-3}$ are shown in Figure 6 for the key parameters (the full contours are shown in Figure B2). We obtain generally unbiased constraints on the cosmological and bias parameters when using the full statistics combination, with $\Omega_{\rm m}$, σ_8, and b_1 all within 1σ from the true values. The trends for the statistics generally follow those seen in the cosmic variance test set (§4.1 and Figure 3), with $P_{gm}(k)$ breaking the degeneracy between σ_8 and b_1, and $B_{ggg}(k_1, k_2, k_3)$ improving constraints significantly for the bias parameters and moderately for the cosmological parameters. For the SHAMe data, we obtain a 34% improvement in precision on $\Omega_{\rm m}$, a 4× improvement on σ_8, and a 4.5× improvement on b_1 when including $P_{gm}(k)$ compared to $P_{gg}(k)$ alone; additionally including $B_{ggg}(k_1, k_2, k_3)$ results in 12%, 11%, and 70% improvements on $\Omega_{\rm m}$, σ_8, and b_1 respectively compared to $P_{gg}(k) + P_{gm}(k)$.

We do see moderate biases in the values of b_2 and b_{s^2} when using the full statistics combination, at the 2.6σ and 1.9σ levels, while these were on average accurately constrainted in the cosmic variance test set. **However, the probabilistic bias approach we used for measuring these has large errors when it comes to these higher-order bias parameters, so results within a few σ are not particularly concerning. That said, some care should be taken with analyses that rely on highly accurate bias parameter inference.** *ksf:* TODO include errors from prob. bias approach; update this section when done!

We show the results for the full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ on the SHAMe samples with a range of number densities in Figure 7. We find that the constraints on the cosmological parameters are robust to the noise level across this range. The constraints on the bias parameter b_1 do lose precision at lower number densities; this could be related to degeneracies with the noise parameters. We note that the noise parameters themselves are poorly constrained in all cases, as seen for the fiducial number density

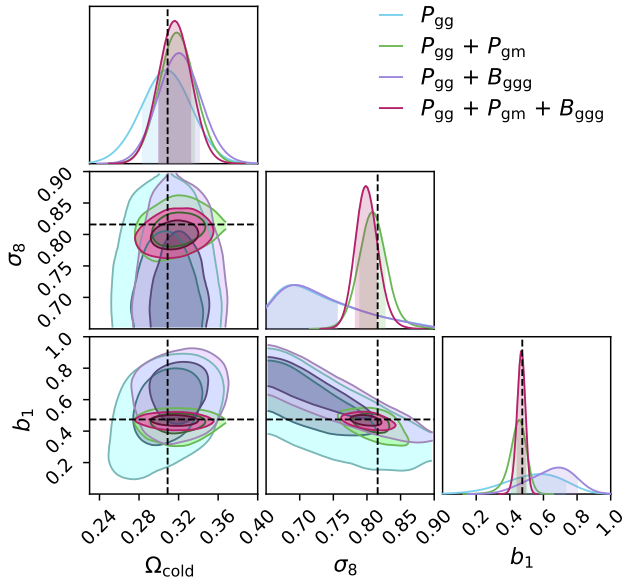


Figure 6. Posteriors for the key parameters Ω_m, σ_8, and b_1 for our out-of-distribution SHAMe mock with fiducial number density $\bar{n} = 2.2 \times 10^{-4}\,(h^{-1}\mathrm{Mpc})^{-3}$, for each of the statistics combinations considered. The recovered parameters with the full statistics combination are unbiased within 1σ, and the statistics show similar constraining power as in the in-distribution test. The full posterior plot is shown in Fig. B2.

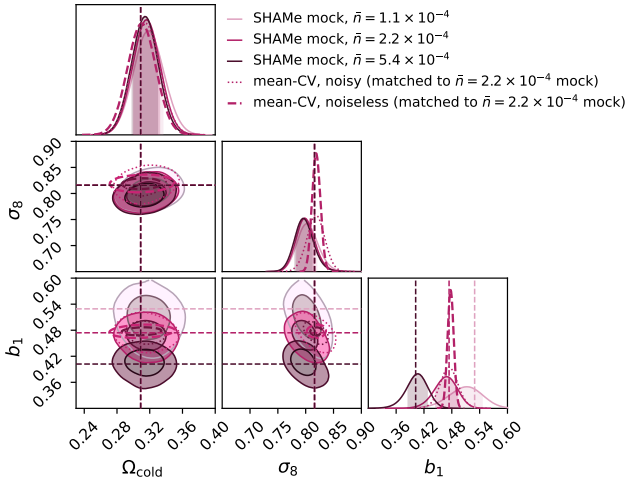


Figure 7. Posteriors for $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ for the key parameters, for varying number density and noise models. The filled contours show the results for the SHAMe OOD mock data with different number densities. The dotted contours show inference on the mean of the cosmic variance test set, which was matched to the fiducial $\bar{n} = 2.2 \times 10^{-4}\,(h^{-1}\mathrm{Mpc})^{-3}$ SHAMe mock; the dashed contour shows that same set but generated without any noise, and inferred assuming noiseless data.

in the full contour plot (Figure B2). However, we check that the noise model is necessary for properly inferring the parameters; assuming noiseless training data results in strong biases in the inference.

We also show the results for inference on the mean data vectors of the cosmic variance sample, which was generated with bias and noise parameters matched to the $\bar{n} = 2.2{\times}10^{-4}\,(h^{-1}\mathrm{Mpc})^{-3}$ SHAMe sample. **The posteriors for these matched in-distribution and out-of-distribution tests is quite similar, with comparable precision on Ω_m and σ_8; the in-distribution mock shows moderately higher precision on b_1. This demonstrates that our cosmic variance sample matched to the SHAMe mock is a good representation of the information content of a sample with those properties** ***dsj:*** **what properties?. The matched in-distribution mock does show an unbiased constraint on σ_8, while the SHAMe mock displays a 1σ bias. This is likely because the inference on the mean data vector approach used for the in-distribution case averages out cosmic variance, whereas for the SHAMe mock we have a single realizaton; though it uses the fixed-and-paired technique, we do not expect it to be immune to cosmic variance effects.** *ksf:* TODO test with matched phases, update this section when done!

We additionally perform a test in which we generate a cosmic variance test set with the SHAMe bias parameters for the fiducial number density sample, but no noise. We construct a noiseless training set, with the same bias fields but an independent nested LH that varies only the 4 bias parameters, and train a new SBI model. When applying this model to the noiseless test set, we find that the constraints are consistent with the noisy case for Ω_m, but improve signficantly for σ_8 and b_1. This demonstrates the constraining power lost to noise for the number density considered, even with perfect knowledge of the noise model.

These tests demonstrate the generalizability of our forward model to realistic galaxy populations constructed with completely different underlying models.

4.3 Scale dependence

We investigate the dependence of our results on the maximum k scale included in our inference, for each of the statistics. We keep the maximum scale for $P_{gg}(k)$ fixed to $k_{max} = 0.37\,h\mathrm{Mpc}^{-1}$, slightly below the Nyquist frequency of the box. We expect the galaxy-galaxy power spectrum to be accurate down to this scale: there the `map2map` emulator of N-body displacements is shown to achieve significantly sub-percent accuracy (~0.2%) (Jamieson et al. 2023), and Pellejero Ibanez et al. (2023a) showed that `map2map` combined with the hybrid Lagrangian bias expansion to second order achieves 1-2% precision to $k_{max} = 0.6\,h\mathrm{Mpc}^{-1}$. (Previous works using the hybrid bias expansion with a power spectrum emulator also find sufficient accuracy down to $k_{max} = 0.6\,h\mathrm{Mpc}^{-1}$ (Hadzhiyska et al. 2021; Zennaro et al. 2023).)*mpi:* parenthesis within parenthesis? I don't think Mr. MNRAS Smith will like this

To assess the accuracy of our results, we use the figure of bias, defined as

$$\mathrm{FoB} = \frac{1}{\chi^2_{\mathrm{lim}}}\sqrt{(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_{\mathrm{true}})^T C^{-1} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_{\mathrm{true}})}\,, \tag{9}$$

where $\hat{\boldsymbol{\theta}}$ are the predicted values of the parameters, $\boldsymbol{\theta}_{\mathrm{true}}$ are the true values, and C is the covariance matrix, which is estimated from the posterior samples. The quantity χ^2_{lim} scales the FoB by the expected χ^2 for the relevant number of dimensions and σ value. The 1σ expectation for three dimensions is $\chi^2_{\nu=3}(0.68) \approx 1.88$, which we use for the parameter subspace $(\Omega_m, \sigma_8, b_1)$; in this case we also compute C only across this subspace. For one dimension this becomes $\chi^2_{\nu=1}(0.68) = 1$, which reduces to the absolute error in terms of the standard deviation, $\mathrm{FoB}_{1D} = |\hat{\theta} - \theta_{\mathrm{true}}|/\sigma_\theta$.

To assess the precision, we use the figure of merit (FoM), defined

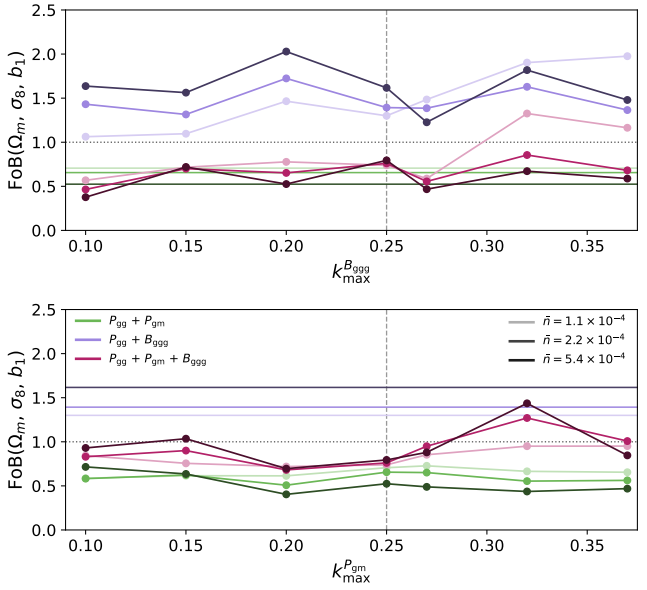


Figure 8. Scale dependence of the figure of bias (FoB) for the three key parameters combined for the SHAMe out-of-distribution mock, with varying number density. The top panel shows the FoB as a function of the $k_{\rm max}$ used for the bispectrum, fixing to our fiducial $k_{\rm max} = 0.25$ for $P_{gm}(k)$; the bottom panel shows the FoB as a function of the $k_{\rm max}$ used for $P_{gm}(k)$, fixing to our fiducial $k_{\rm max} = 0.25$ for the bispectrum. In both cases, the FoB for the full statistics combination remains around or below 1σ until $k_{\rm max} = 0.25$, after which the inference shows a bias; we thus use $k_{\rm max}^{P_{gm}} = k_{\rm max}^{B_{ggg}} = 0.25$ for our fiducial inference. *ksf:* increase legend size *dsj:* units for horizontal axes and $\bar{n}$ values.

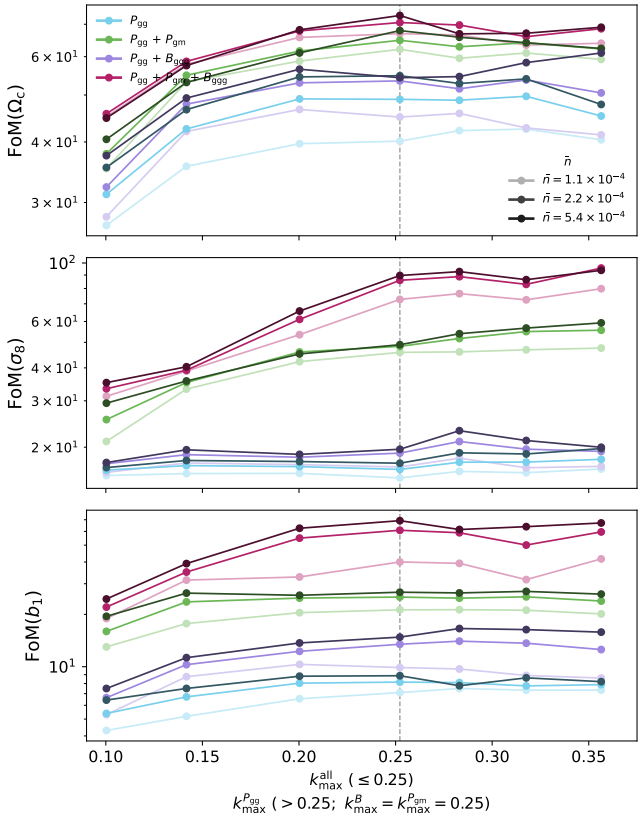


Figure 9. Scale dependence of the figure of merit (FoM) for the three key parameters for the SHAMe out-of-distribution mock, with varying number density. The $k_{\rm max}$ displayed is defined as follows: for $k_{\rm max} < 0.25$, $k_{\rm max} = k_{\rm max}^{\rm P_{gg}} = k_{\rm max}^{\rm P_{gm}} = k_{\rm max}^{\rm B_{ggg}}$; for $k_{\rm max} > 0.25$, we fix $k_{\rm max}^{\rm P_{gm}} = k_{\rm max}^{\rm B_{ggg}} = 0.25$ and only vary $k_{\rm max} = k_{\rm max}^{\rm P_{gg}}$. Precision generally increases with scale, especially from $k_{\rm max} = 0.1$ to $k_{\rm max} = 0.25$, with additional gains for σ_8 when extending to $k_{\rm max}^{P_{gg}} = 0.37$. *ksf:* make legends, axis labels bigger *dsj:* units missing here too

as

$$\mathrm{FoM} = \frac{1}{\sqrt{\det C}}\,, \tag{10}$$

which is the inverse of the volume of the allowed region, such that a higher FoM indicates tighter constraints. Again we compute C only across the subspace considered; in one dimension, this reduces to $\mathrm{FoM_{1D}} = 1/\sigma_\theta$.

In Figure 8, we show the figure of bias for the SHAMe out-of-distribution mock for the three key parameters $\Omega_{\rm m}$, σ_8, and b_1 combined, as a function of the $k_{\rm max}$ of $B_{ggg}(k_1, k_2, k_3)$ and $P_{gm}(k)$; when varying $k_{\rm max}^{\rm B_{ggg}}$, we fix $k_{\rm max}^{\rm P_{gm}} = 0.25\,h\mathrm{Mpc}^{-1}$, and when varying $k_{\rm max}^{\rm P_{gm}} = 0.25\,h\mathrm{Mpc}^{-1}$ we fix $k_{\rm max}^{\rm B_{ggg}} = 0.25\,h\mathrm{Mpc}^{-1}$. We see that using only $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ results in a bias moderately above 1σ for all number density samples, regardless of $k_{\rm max}$; this is likely largely a result of projection effects, as we have found in earlier tests. When including the galaxy-matter power spectrum, $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$, the inference is accurate for all number densities to 1σ up until $k_{\rm max}^{\rm B_{ggg}} = 0.25\,h\mathrm{Mpc}^{-1}$, where the lower number density samples begin to exhibit a slight bias. This aligns with the findings of Pellejero-Ibáñez et al. (2026) that validated the hybrid bias expansion down to $k_{\rm max} = 0.25\,h\mathrm{Mpc}^{-1}$.

The statistics combination $P_{gg}(k) + P_{gm}(k)$ results in accurate inference for all number densities through $k_{\rm max}^{\rm P_{gm}} = 0.37\,h\mathrm{Mpc}^{-1}$. However, when additionally adding $B_{ggg}(k_1, k_2, k_3)$, the inference becomes biased pushing below $k_{\rm max}^{\rm P_{gm}} = 0.25\,h\mathrm{Mpc}^{-1}$. *ksf:* anything to cite/mention for the cross-power-spectrum accuracy? It is initially odd that pushing $P_{gm}(k)$ to smaller scales results in a bias only when including the $B_{ggg}(k_1, k_2, k_3)$; we speculate that this could be because our noise model does not account for noise in $P_{gm}(k)$, though further investigation is merited. While here we have shown the dependence varying the $k_{\rm max}$ of only one statistic at a time, we also check the grid of combinations for $k_{\rm max}^{\rm B_{ggg}}$ and $k_{\rm max}^{\rm P_{gm}} = 0.25\,h\mathrm{Mpc}^{-1}$, and find $k_{\rm max}^{\rm B_{ggg}} = k_{\rm max}^{\rm P_{gm}} = 0.25\,h\mathrm{Mpc}^{-1}$ to be the optimal combination of limiting scales; pushing either or both to smaller scales results in further bias. We also note that there is no clear trend with number density across the range tested.

mpi: You have a bunch of k-units missing in these paragraphs as well. And in the plots too. We explore the dependence of the figure of merit on the maximum scale included in the inference in Figure 9. We show the FoM for the three key parameters for the SHAMe mocks at each of the three number densities tested, for all statistics combinations. For $k < 0.25$, we apply the same $k_{\rm max}$ for all three statistics; for $k > 0.25$, we fix $k_{\rm max}^{\rm P_{gm}} = k_{\rm max}^{\rm B_{ggg}} = 0.25$ (the fiducial values chosen based on the FoB analysis above) and only include higher k for $P_{gg}(k)$. We first see that higher number density samples consistently result in higher precision constraints, with the increases strongest for b_1, as we saw for $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ in §4.2, though the difference in information gain as a function of maximum scale does not display a clear trend.

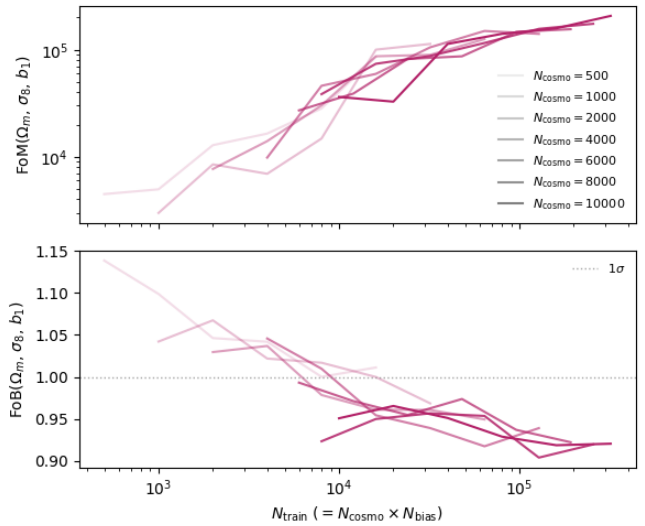


Figure 10. Convergence of the inference with training set size, for $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ on the coverage test set for the three key parameters $\Omega_{\rm m}$, σ_8, and b_1 combined. The top panel shows the figure of merit (FoM), and the bottom panel shows the figure of bias (FoB), normalized by the 68th percentile of the χ^2 distribution for three dimensions (corresponding to the three parameters); the median values over the coverage set are shown. The opacity indicates the number of training cosmologies. Increasing the number of training points continues to increases the precision, with the number of cosmologies and bias parameter sets contributing similar precision. *ksf:* make legends, axis labels bigger *ksf:* note: this plot is with an older base model, with kmax=0.4 and no ensembling; new version is running. also: should we make this with only the inner 500, as in the coverage tests?

We find that for nearly all of the statistics combinations, parameters, and number densities, there is a significant increase in precision from $k = 0.1$ to $k = 0.25$. The exception is $P_{gg}(k)$ alone for σ_8; the information content is already saturated at larger scales. Interestingly, there is not signficant information gain for most statistics combinations and parameters beyond $k_{\rm max} = 0.25$. We find that the strongest increases in the FoM from $k_{\rm max}^{\rm all} = 0.2$ to $k_{\rm max}^{\rm P_{gg}} = 0.37$ is for σ_8; for the fiducial number density $\bar{n} = 2.2 \times 10^{-4}\,(h^{-1}{\rm Mpc})^{-3}$, we find that constraints with $P_{gg}(k) + P_{gm}(k)$ increase by 21% and with $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ by 57%. These are quite significant gains; this demonstrates the value of using the hybrid bias expansion approach in comparison to EFT approaches, which are limited to $k =$~0.1–0.2 *mpi:* references?.

4.4 Dependence on number of cosmologies and bias parameter sets

We explore the convergence of our inference with the number of training data points. We vary the number of cosmologies by selecting only a subset of cosmologies to train the network with, choosing $N_{\rm cosmo} = 500, 1000, 2000, 4000, 6000, 8000, 10000$; the subsets are nested to minimize random variance. To vary the number of bias parameters, we train on each layer of the nested Latin hypercube of bias and noise parameters, resulting in a range of bias parameters per cosmology: $N_{\rm bias} = 1, 2, 4, 8, 16, 32$. For each of these combinations of the number of cosmology and bias parameters, we train a new model using the fiducial hyperparameters for the given statistics combination, and perform inference on the 1,000 mocks of the coverage set.

We compute the figure of merit and figure of bias for the three key parameters combined, taking the median over the coverage set for the full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$. Figure 10 shows this as a function of the total number of training points $N_{\rm cosmo} \times N_{\rm bias}$, displayed as separate lines for each choice of $N_{\rm cosmo}$. We see that the FoB reduces sharply until $N_{\rm train} = 10,000$ where it crosses FoB $= 1\sigma$ but then more-or-less plateaus, indicating that above a few tens of thousands of training points the training data is sufficient for unbiased inference. Similarly, the FoM increases with increasing number of training points, steeply until $N_{\rm train} = 10,000$ and then more moderately. However, the FoM does not plateau even at 320,000 training points, indicating that the inference is still not converged and the precision would likely continue to improve with additional training data. **Notably, the slope is quite shallow, indicating that many more training points may be needed for continued gains.**

We also find that there is not a strong trend with the number of training cosmologies, as evidenced by the lines for the different cosmologies largely overlapping; the trend is instead explained by the total number of training points. With a smaller number of training data points (below $N_{\rm train} = 20,000$), it is more important to have more cosmologies rather than more bias parameter sets, shown by the spread in the FoB around that value; however, beyond $N_{\rm cosmo} = 2000$, it is sufficient to add more bias parameter sets per cosmology. This somewhat surprising finding is encouraging given that generating training data at many cosmologies is more expensive than populating each cosmology with many bias parameter sets. Previous SBI analyses already use around this number of training cosmologies with full N-body codes (without the need for emulators, e.g. the Quijote Latin hypercube)*mpi:* references?; our result shows that this may be sufficient in terms of cosmologies, but that additional constraining power and robustness could be gained by populating these with many more galaxy formation parameter sets per cosmology.

In Figure 11, we show the FoM and FoB as a function of the training set size for all the statistics combinations, for each key parameter individually. Note that for a Gaussian sample, we would expect FoB ≈ 0.8; our FoB stays slightly higher than this for all the parameters, bottoming out at 0.85-1.0, possibly as a result of non-Gaussian behavior. For $P_{gg}(k)$ and $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$, we see that a small number of training cosmologies already saturates the precision for σ_8, though around $N_{\rm train} = 20,000$ are needed to reach a plateau in the FoB below 1σ. For $\Omega_{\rm m}$ and b_1, we see moderate improvement in precision as we increase the training set size across the range explored, with b_1 beginning to plateau around $N_{\rm train} = 100,000$. The statistics combinations $P_{gg}(k) + P_{gm}(k)$ and $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ show steep improvements in both precision and accuracy until $N_{\rm train} = 10,000$ for all parameters, and then improve more moderately. Similarly as we observed in Figure 10, we see that by the maximum number of training points $N_{\rm train} = 320,000$, the FoM continues to rise, for most of the individual parameters and statistics combinations. This indicates that additional training data will likely continue to improve the inference and we have not yet extracted the full information content the statistics. We also again find that for all parameters, increasing the number of bias parameter sets per cosmology results in similar FoM and FoB values as increasing the number of cosmologies, beyond a few thousand cosmologies.

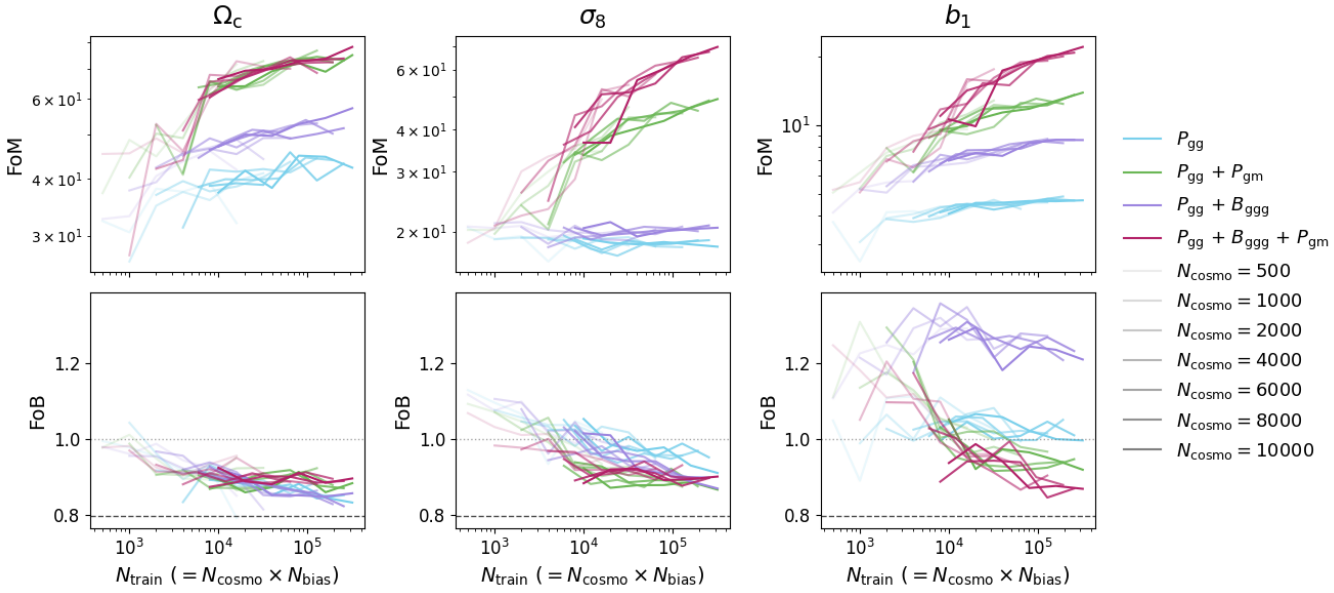


Figure 11. Convergence of the inference with training set size, for all the statistics combinations on the coverage test set for the three key parameters Ω_m, σ_8, and b_1 individually. The top row shows the figure of merit (FoM) and the bottom row shows the figure of bias (FoB); the median values over the coverage set are shown. The opacity indicates the number of training cosmologies, and the dashed line indicates the expectation for an unbiased Gaussian distribution. The FoM increases with training set size for most statistics and parameters, and most reach an FoB around or below 1σ. *ksf:* note: this plot is with an older base model, with kmax=0.4 and no ensembling; new version is running. also: should we make this with only the inner 500, as in the coverage tests?

5 DISCUSSION

The results of this analysis outline the successes and challenges of SBI with the hybrid Lagrangian bias expansion and emulated N-body displacements. While we showed that our model is able to provide robust inference on cosmology, model extensions and adjacent analyses will add to our understanding of the information content in galaxy clustering and optimal approaches to extract it.

We have used Lagrangian bias expansion to second order; it is possible that extending into the third order may be necessary for more accurate modeling in the small-scale regime *ksf:* citation for this / more to say?. For the stochastic noise, we have used a noise model based on a Gaussian field coupled to the bias fields following Rubira & Schmidt (2025), to first order in the noise field. This is a more sophisticated model than a standard Gaussian field, or parametric noise models specific to each statistic, and we showed it to be valid for $B_{ggg}(k_1, k_2, k_3)$. However, this still does not capture the noise in $P_{gm}(k)$ and may not be suffient for other high-order statistics. Extending this noise model to higher order (at the expense of adding additional parameters) or testing other noise models may be important for ensuring the most robust inference.

The `muchísimocks` suite uses the original `map2map` emulator, which has been largely validated for our use case. However, there is an updated version of `map2map` that includes redshift dependence and models the velocity field as the derivative of the position field, and is fully differentiable (Jamieson et al. 2024).[4]. This would allow for inference across a range of redshifts, and the opens the door to inference methods that utilize the differentiability property. It may also address some remaining issues with the original forward model, such as the moderate bias found by Pellejero Ibanez et al. (2023a) for the bispectrum at intermediate scales *mpi:* I'm confused, what bias? I don't remember measuring the bispectrum. The emulator also has not been robustly validated for the whole range of higher order statistics or the full tracer field; improving the displacement emulator and further validating it will be key for future analyses. *ksf:* updated this paragraph

Bairagi et al. (2025) explored the question of the required number of simulations for cosmological inference, using a large suite of 32,768 N-body simulations called the Big Sobol Sequence. Using neural posterior estimation on the nonlinear dark matter power spectrum down to $k_{max} = 0.8$, they found that around 4,000 simulations were required to saturate the information content, signifcantly more than the 2,000-simulation Quijote suite. This stands somewhat in contrast to our finding, that on the one hand 2,000 cosmologies provide comparable results to 10,000 given an equal total number of simulations (including multiple galaxy bias parameter sets), and on the other that even 320,000 total training data points does not seem to saturate the precision. However, our convergence findings are not directly comparable, as their work analyzes a matter field statistic and does not sample galaxy bias parameters, so we would expect that their setup requires fewer simulations. Tucci & Schmidt (2024) also analyze the convergence of their EFT-based SBI approach with the number of simulations, finding that at fixed cosmology 10^5 simulations are required to converge in the inference of the bias parameters. In the free-cosmology case with one cosmological parameter, the inference is slower to converge, and is not clearly fully converged at their maximum 5×10^5 simulations; this is in line with our results that upwards of 10^5 simulations are required for convergence. *dsj:* Ours also doesn't converage, and we do not know how many we would need to converge, but it looks like $\gg 10^5$. "Upwards of 10^5" doesn't make this clear.

Our finding that even 320,000 training data points do not result in converged inference are important in interpreting our

[4] https://github.com/dsjamieson/jax_nbody_emulator

results and guiding future SBI approaches. First, this means it is possible that our estimations of the precision gain from additional statistics are not stable. That said, we see a consistent or increasing difference in precision when including additional statistics as a function of the number of training points, so the values presented here can be taken as conservative estimates. Regarding future implications, we see that given the slope of the convergence with the training set size, it is possible that an order of magnitude or more simulations may be required. While this is possible with the `muchísimocks` framework to a point given its speed, especially given that we find that additional bias parameter sets provide equivalent improvement to additional cosmologies at a much lower cost, we are still limited by the cost of computing summary statistics, and model training time increases with training set size. This motivates exploring other ways of achieving convergence with fewer simulations, such as adaptive sampling of the parameter space, reducing the dimensionality of the parameter space, or sequential SBI. Further work with large simulation suites in various regimes and different modeling choices is necessary to provide a complete picture of the convergence with the number of simulations **and guide future analyses**.

The `muchísimocks` approach has the potential to be expanded to answer further questions about the information content in galaxy clustering and the requirements for SBI. For instance, we could straightforwardly generate larger volumes with this approach, as current survey volumes are easily outpacing our ability to model these volumes with traditional simulations. **We could also incorporate velocity information to generate redshift-space fields and investigate the information content in redshift-space distortions.** With modifications to the forward model, beyond-ΛCDM cosmologies could be explored to understand the sensitivity of various statistics to alternative cosmological models.

A key potential extension is to explore other statistics of the tracer fields to understand their information content. Optimal summary statistics that capture the information content of the full field could be learned, to explore the maximal information in cosmological fields. As `muchísimocks` also includes a forward model of the velocity field, this suite could be useful for exploring velocity reconstruction approaches. One could also investigate the effect of more restrictive priors on the Lagrangian bias parameters, encoding our knowledge specific galaxy samples.

The SBI approach with the `muchísimocks` model that we have demonstrated could be applied to more realistic data, and eventually to real data to perform a cosmological analysis. This analysis has been entirely in real-space, and a redshift-space analysis is forthcoming. *ksf:* this is repeated a bit from above now; prefer one place or both, and also should we say forthcoming or nah? Survey realism, including systematics, could be applied to the fields to understand the information content in more realistic settings and ensure the inference remains robust. Finally, an analysis on survey data such as BOSS or DESI has the potential to provide competitive and complementary measurements of cosmological parameters.

6 SUMMARY AND CONCLUSIONS

In this work, we have presented a forward model for the full galaxy tracer field that balances robustness and speed, making it ideal for simulation-based inference approaches to cosmological inference. The model uses the hybrid Lagrangian bias expansion to second order, based on emulated particle displacement fields from the `map2map` emulator of Jamieson et al. (2023) trained on the Quijote mocks, to generate sets of bias fields that can be weighted and summed to produce tracer fields. We also include a full-field noise model of Rubira & Schmidt (2025) based on a Gaussian field coupled to the bias fields, which we find to be critical for unbiased inference. With this forward model, a $(1h^{-1}\mathrm{Gpc})^3$ mock can be generated at any cosmology in around five minutes. We publicly release a suite of these bias fields, called `muchísimocks`, at $z = 0$ in both real- and redshift-space. Our main mock set consists of a Latin hypercube of 10,000 different cosmologies; we also release an independent 1,000-cosmology Latin hypercube "coverage" test set, and a 1,000-realization "cosmic variance" set at fixed cosmology with varying seeds.

We perform simulation-based inference using use neural posterior estimation. We generate training data from our forward model, recombining each bias field set at a given cosmology with up to 32 bias and noise parameter sets, for a total of 320,000 traning mocks. The summary statistics considered are the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter cross-spectrum $P_{gm}(k)$, and the galaxy bispectrum $B_{ggg}(k_1, k_2, k_3)$. In addition to the in-distribution tests sets described above, we apply our model to a realistic out-of-distribution mock built with the extended Subhalo Abundance Matching (SHAMe) model on a high-resolution BACCO simulation. **This is the first cosmological inference framework based on the hybrid bias expansion involving higher-order clustering statistics, and we demonstrate that it robustly infers unbiased and precise cosmological parameters.**

Our specific findings are the following:

- Inference with $P_{gg}(k)$ alone suffers from strong projection effects, resulting in biased σ_8 even for the in-distribution test sets; this holds even when additionally including $B_{ggg}(k_1, k_2, k_3)$. Including $P_{gm}(k)$, which is related to the galaxy–galaxy lensing observable, results in unbiased inference of Ω_m, σ_8, and b_1, as it breaks parameter degeneries and ameliorates projection effects (see Figs. 3 and 4).
- We recover unbiased cosmological parameters on the out-of-distribution SHAMe mock, validating the robustness of our model. This holds down to scales of $k_\mathrm{max} = 0.35$ for the galaxy–galaxy power spectrum, and $k_\mathrm{max} = 0.25$ for the galaxy–matter cross spectrum and the galaxy bispectrum (see Figs. 6 and 8).
- Increasing the number of bias parameter sets applied to each cosmology achieves a similar precision and accuracy as increasing the number of cosmologies, for the same total number of points, beyond a minimum of around 2,000 cosmologies (see Fig. 11). This could be a very helpful result, as augmenting with more galaxy bias combinations is far faster than generating mocks at new cosmologies.
- For $P_{gg}(k)$ alone, around 20,000 training mocks are required for inference to converge and information content to saturate; for the full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$, the precision continues to increase even with 320,000 training points, indicating that additional mocks (or improved SBI procedures) are needed to saturate the information in those statistics (see Fig. 10).

The `muchísimocks` suite will be publicly available, and is useful for various analyses exploring the information content in galaxy clustering. With its large number of mocks and robust method for modeling the galaxy–dark matter connection, it is ideal for testing approaches to simulation-based inference. The approach could be extended to generate mocks exploring a range of science questions and targeted for specific tracer populations and surveys, paving the way for robust and flexible inference from galaxy clustering.

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SOFTWARE AND LLM USAGE

The following software were used in this work: *ksf:* TODO fill in software In this work, large language models (LLMs) were used to make code modifications, write basic code such as plotting functions, support background research, and provide basic LaTeXsupport. Core code and all writing did not involve the use of LLMs.

DATA AVAILABILITY

The `muchísimocks` suite and the code used in this paper will be made publicly available at `thisurl`.

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APPENDIX A: VALIDATION OF INFERENCE ON MEAN DATA VECTOR

In this analysis, we have shown results of the SBI pipeline with the `muchísimocks` forward model on the "cosmic variance" test set, which contains 1,000 realizations run with different seeds at the same cosmology, bias, and noise parameters. Specifically, we took the mean of the 1,000 data vectors computed on each of these mocks individually, and used that mean data vector as the input to the SBI model. In this section, we investigate this analysis choice and its implications.

We first compute the statistics on all 1,000 mock realizations, and perform the inference on each individually with our SBI pipeline. We obtain a posterior for each of the 1,000 realizations. To visualize the "typical" posterior, we compute the mean-of-the-means of the 1,0000 posteriors, and shift each posterior such that it is centered on that value; we then pool the posteriors, drawing 500 samples from each.

We then compute the mean of the statistics of the 1,000 mocks, and perform the inference on that mean data vector. This mean data vector has the benefit of reducing the effect of cosmic variance on the statistic, but brings the complication of having a different covariance than the data the SBI model was trained on (namely, much lower variance). In standard MCMC approaches, this would be handled by appropriately scaling the covariance matrix, but in the context of SBI, the covariance is learned by the model from the training data and is not given explicitly. Thus it is important to understand the behavior of the model when we use the mean data vector for inference.

Figure A1 shows the result of these ways of analyzing the cosmic variance test set, all for the combined statistic $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$. The grey contours show posteriors of 10 randomly selected individual mocks; the brown contour shows the recentered-pooled posteriors (over all 1,000 mocks); and the magenta contour shows the inference performed on the mean of the 1,000 data vectors. The brown dot and vertical lines show the mean of the means of the 1,000 individual posteriors.

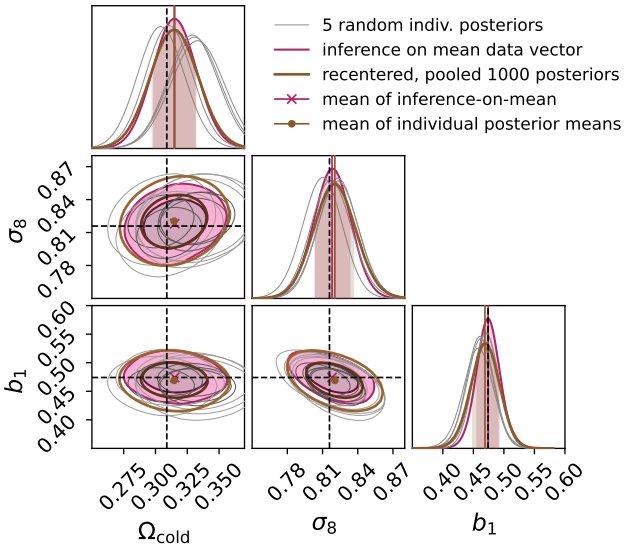


Figure A1. Validation of inference on the mean of the data vectors for the cosmic variance test set, for the full $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ combination. The grey contours show posteriors from inference performed on statistics of 5 randomly selected realizations from the test set; the brown contours are pooled samples from each of the 1,000 realizations, 500 of each recentered to the mean of the means; the magenta contours show inference performed on the mean of 1,000 data vectors. The mean of the inference-on-the-mean (magenta vertical line and X marker) nearly exactly agrees with the mean of the 1,000 individual posterior means (brown vertical line and dot marker). The shape and width of the posteriors are also broadly consistent.

We see that the inference on the mean produces a posterior very close to that of the recentered-pooled posterior. The means, as well as the MAPs, for these two results are within ∼1% for all parameters; both are within 0.5σ (2%) from the true values. The predicted uncertainty regions are also very similar: the symmetrized inner 68% width agrees to 0.2% for $\Omega_{\rm m}$, 4% for σ_8, and 4% for b_1. We check that the recentered-pooled posterior width agrees with the mean of the widths of the 1,000 individual posteriors, to better than 0.5%. The inference-on-the-mean does generally produce slightly tighter constraints than the average of the individual posteriors, interpreted as mild overconfidence. However, the mean data vector should be less affected by prior effects than individual realizations with summary statistics towards the edges of the distribution. *ksf:* i think we still don't know exactly how to interpret this; so i am not sure how to tie these last couple sentences together.

This analysis indicates that despite the lower variance of the mean data vector, the SBI pipeline produces a posterior that has the typical width of a posterior estimated with the noisier data vector, as that was closer to the noise levels of data it was trained on. We thus use the inference on the mean data vector throughout the paper to be able to show the typical result of the inference, with the effect of cosmic variance (which would cause a bias in any given individual sample) removed. We emphasize that there is not a clear statistical interpre-

tation of this inference-on-the-mean for simulation-based inference, and the stability of the comparison to the individual posteriors varies somewhat across models. However, the results empirically demonstrate that this is a useful way to validate an SBI pipeline with a much lower computational burden compared to sampling many individual posteriors.

APPENDIX B: FULL POSTERIOR CONTOURS

We show the full contour plots for all statistics combinations considered, for all of the varied parameters: the two cosmological parameters, four bias parameters, and five noise parameters. Figure B1 shows the posteriors inferred from the mean of the data vectors of the 1,000-mock cosmic variance test set. Figure 6 shows the inference on the out-of-distribution SHAMe mock at fiducial number density, $\bar{n} = 2.2 \times 10^{-4}\,(h^{-1}\mathrm{Mpc})^{-3}$.

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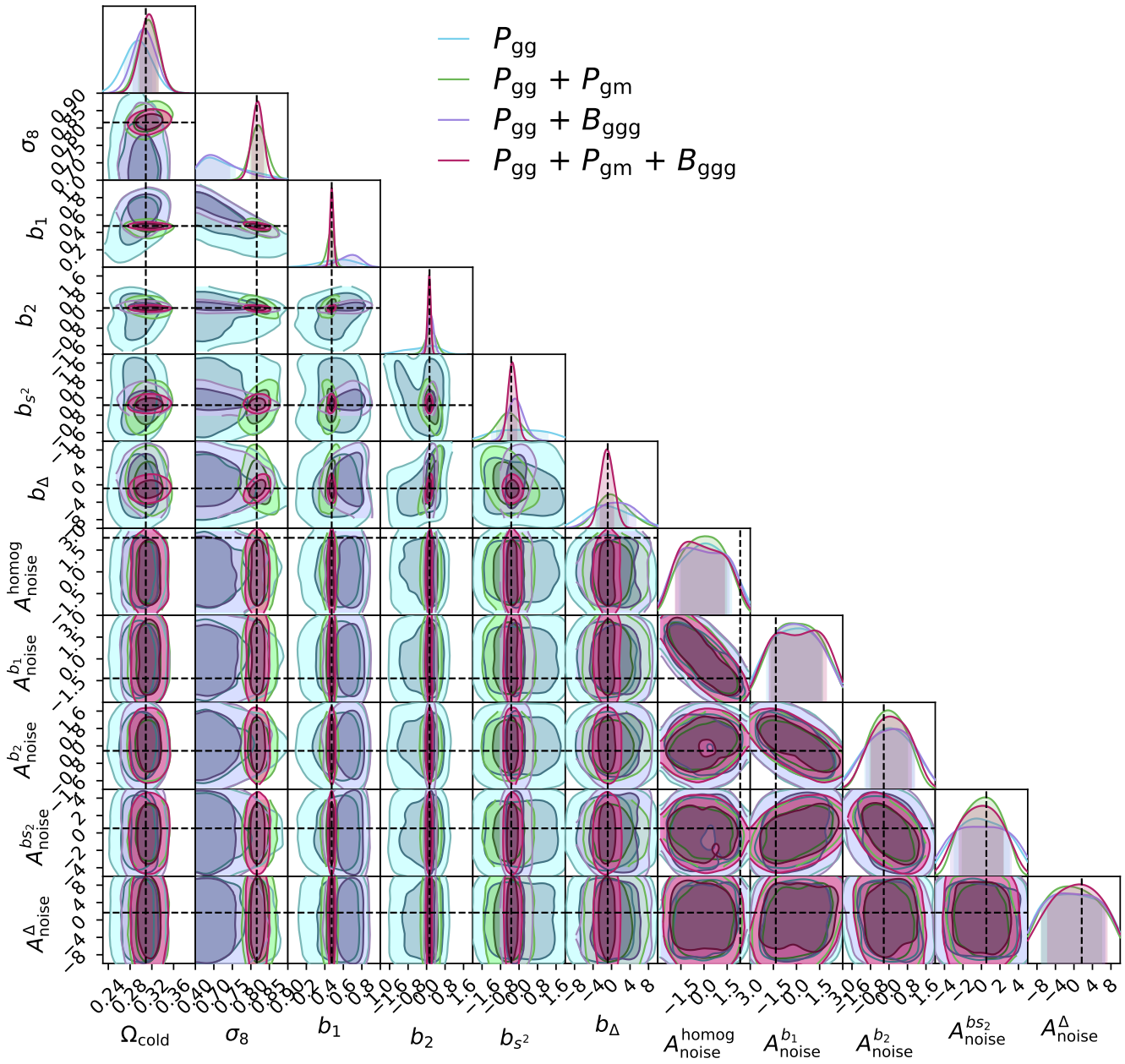


Figure B1. Full posterior contours for the cosmic variance test set, inferred on the mean of the data vectors across this set.

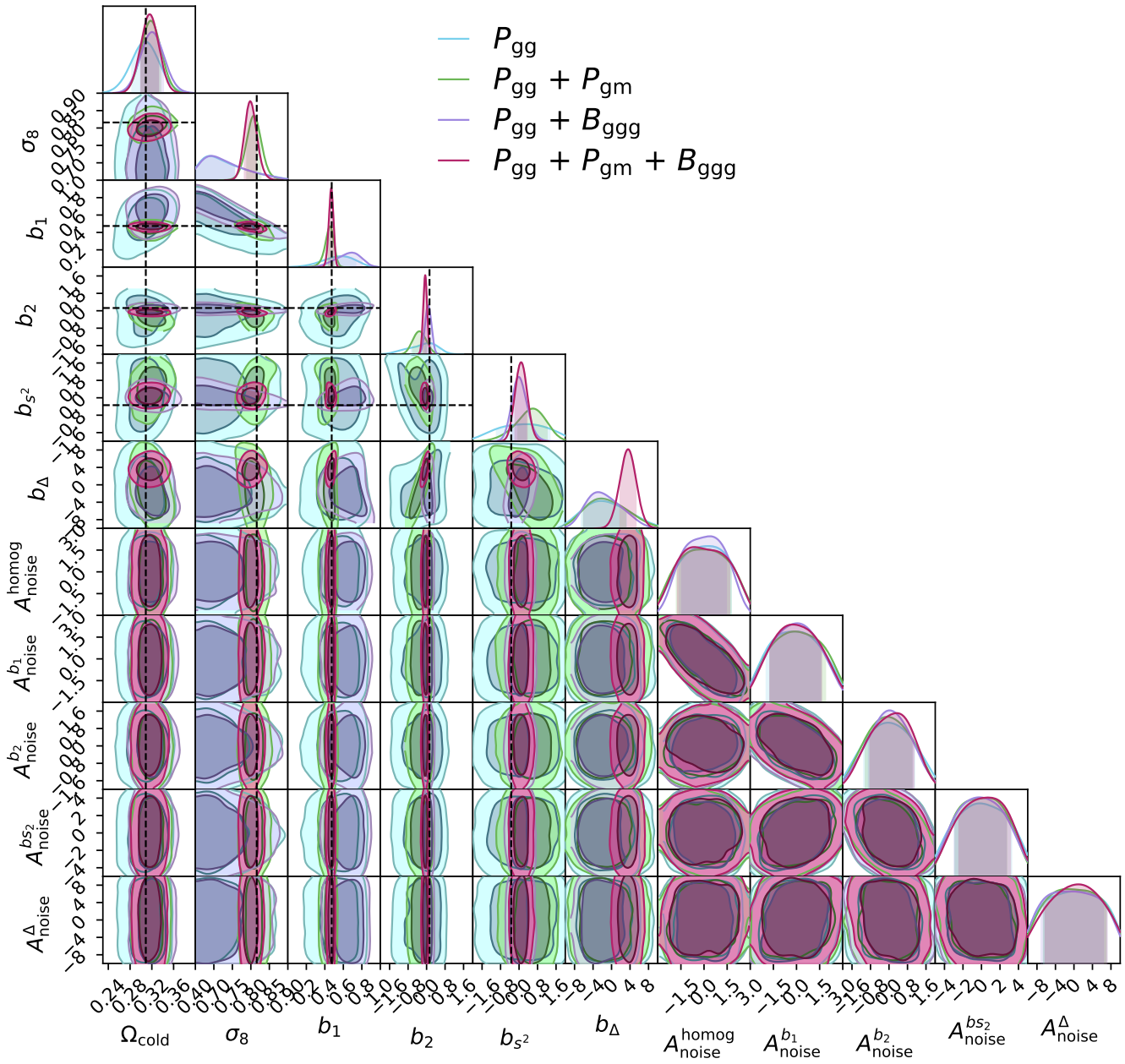


Figure B2. Full posterior contours for the SHAMe out-of-distribution test set with $\bar{n} = 2.2 \times 10^{-4}\,(h^{-1}\mathrm{Mpc})^{-3}$.